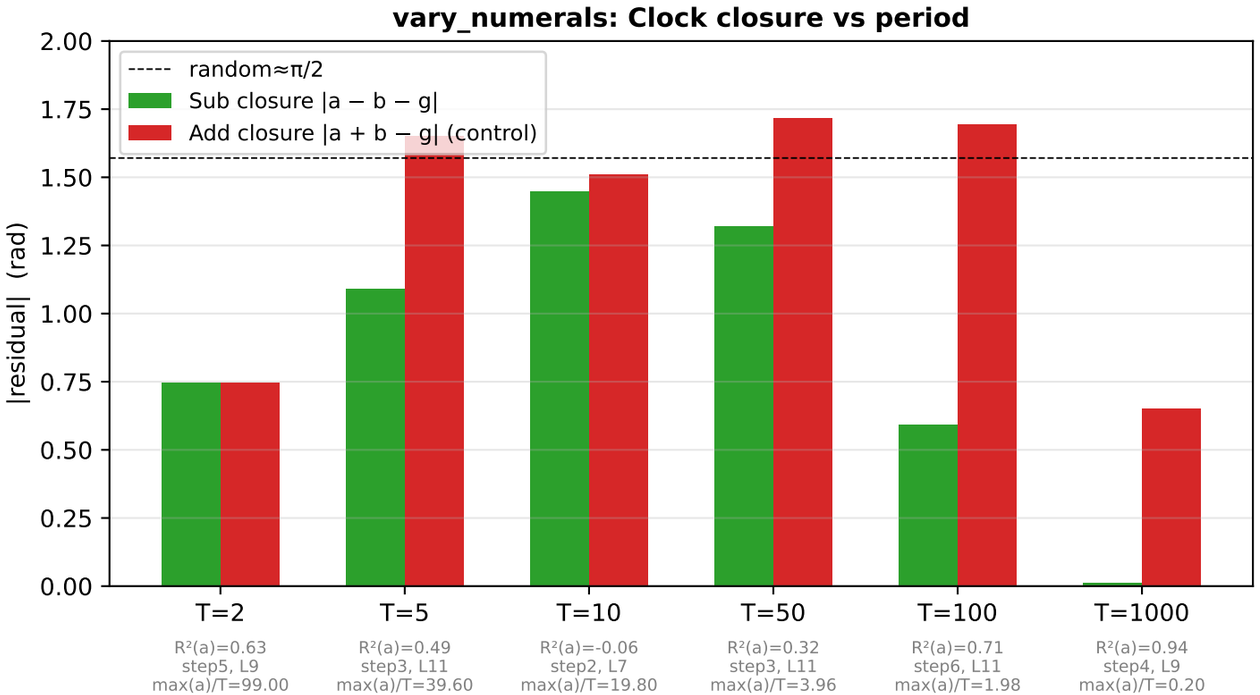
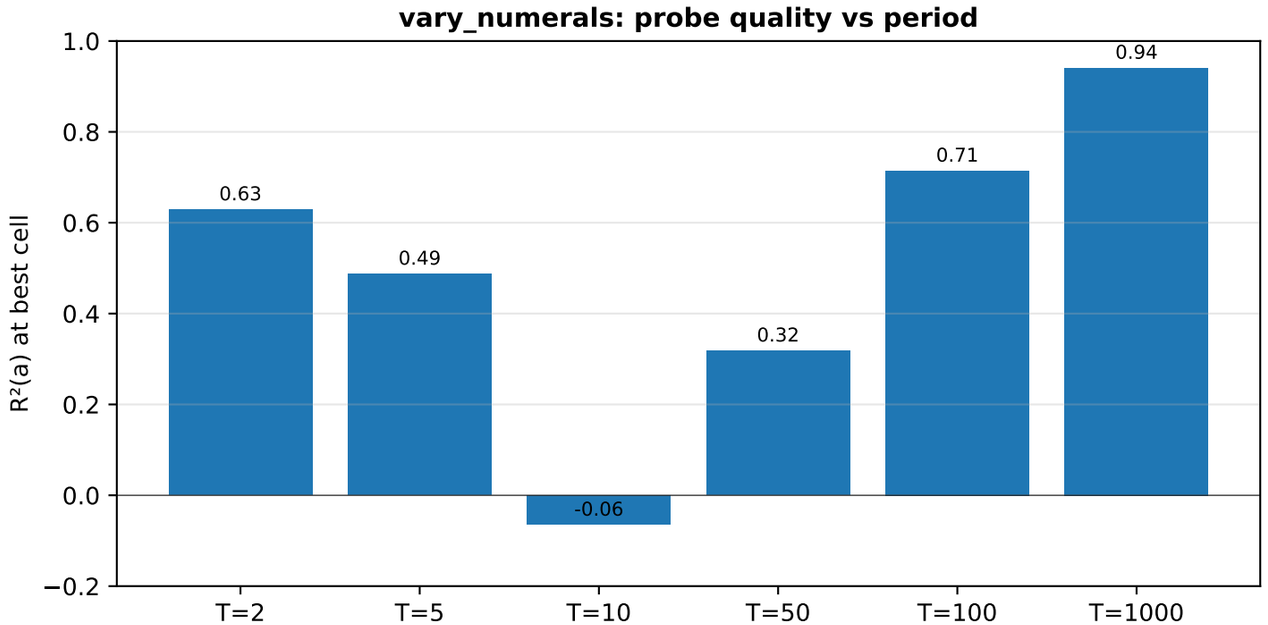
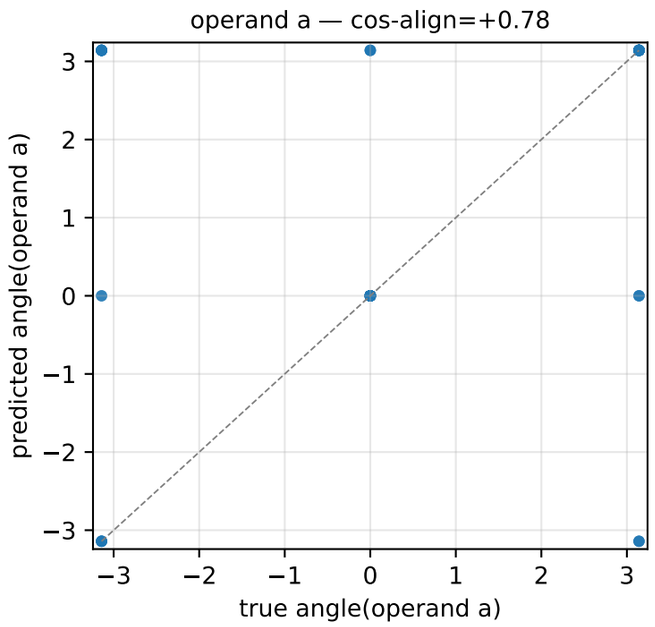
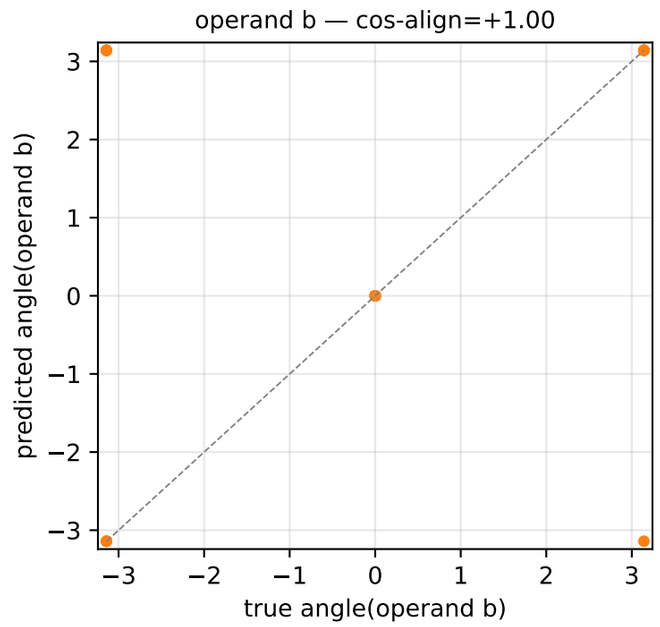
Clock test on CODI-GPT-2 latent residuals: vary_numerals

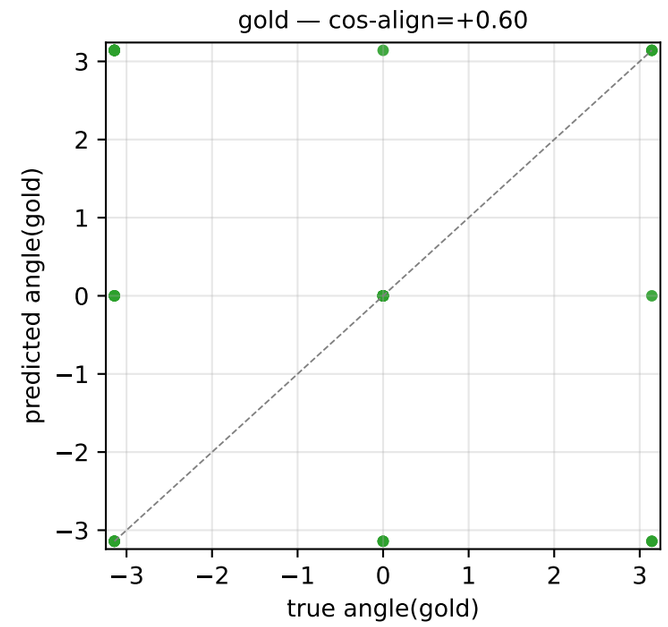


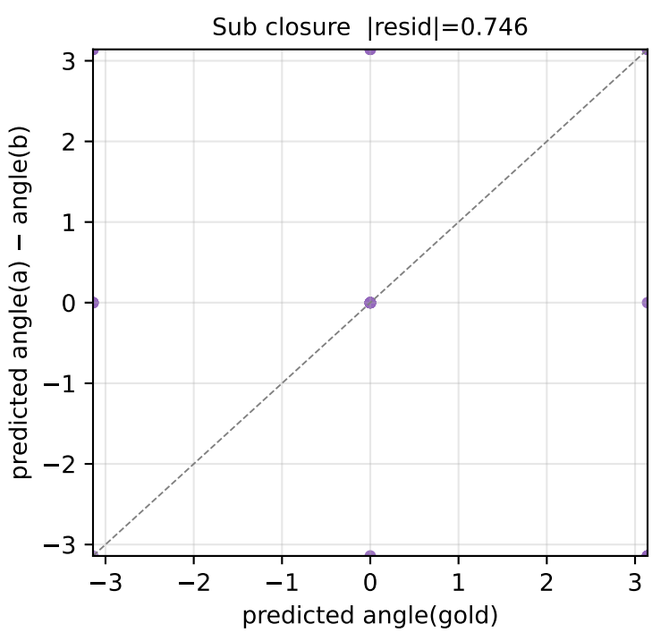


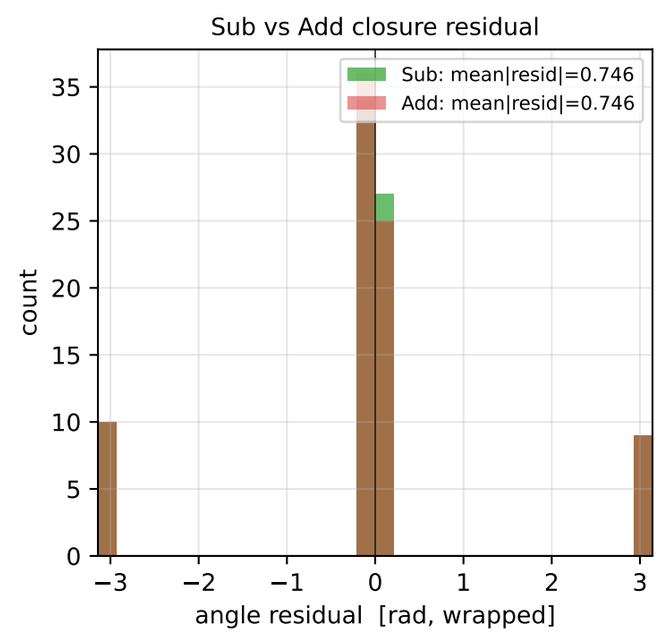
vary_numerals — Clock test at T=2





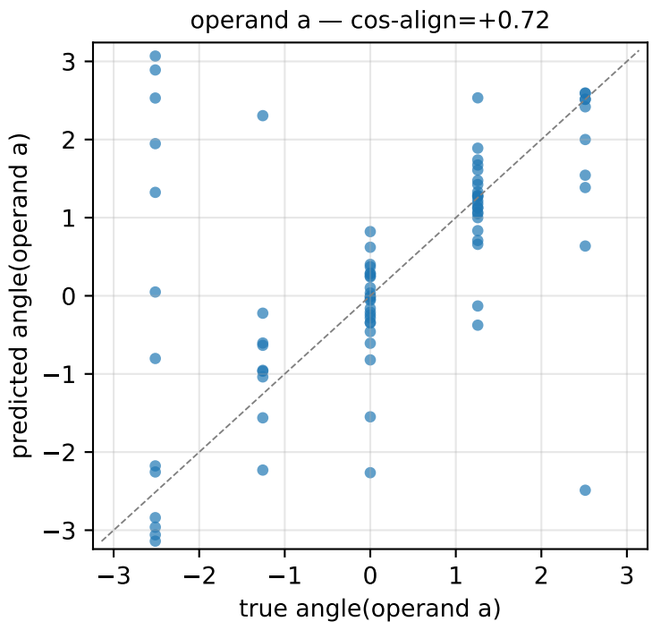


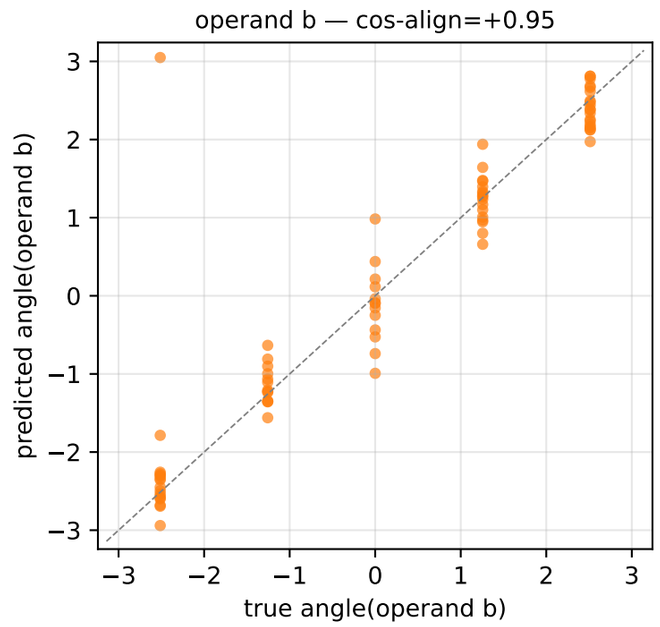


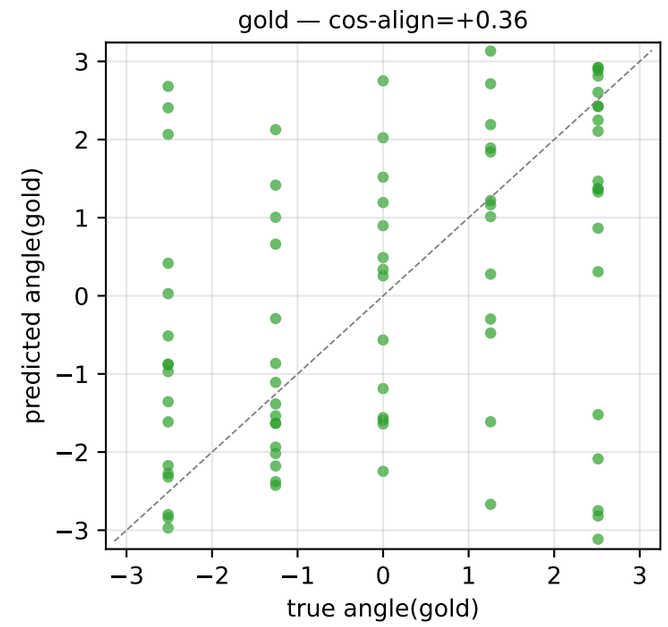


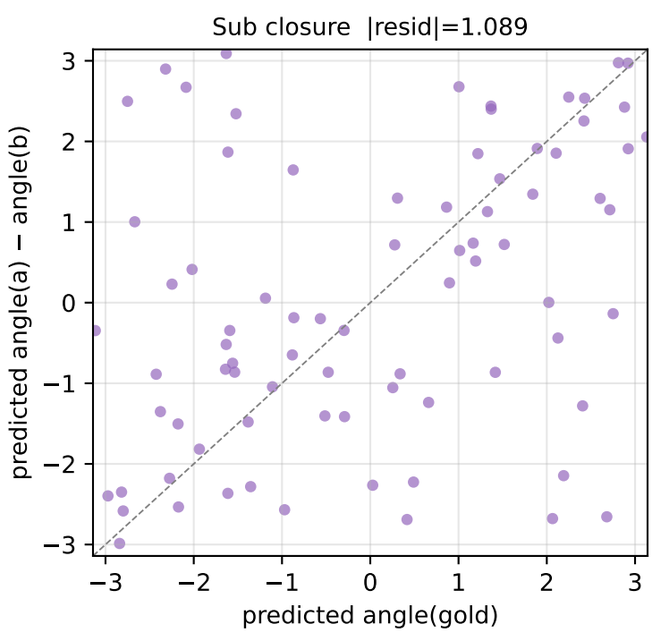

```
Period T = 2
Best cell:  step=5, layer=9
R²(a) = +0.629
cos-align(pred angle, true angle) = +0.775
max(operand)/T = 99.00
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.746 rad
|angle(a) + angle(b) − angle(gold)|  =  0.746 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

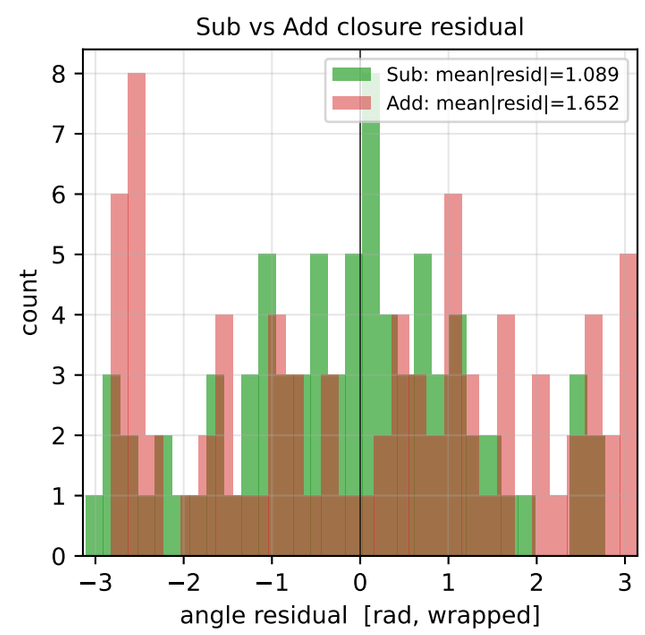
vary_numerals — Clock test at T=5





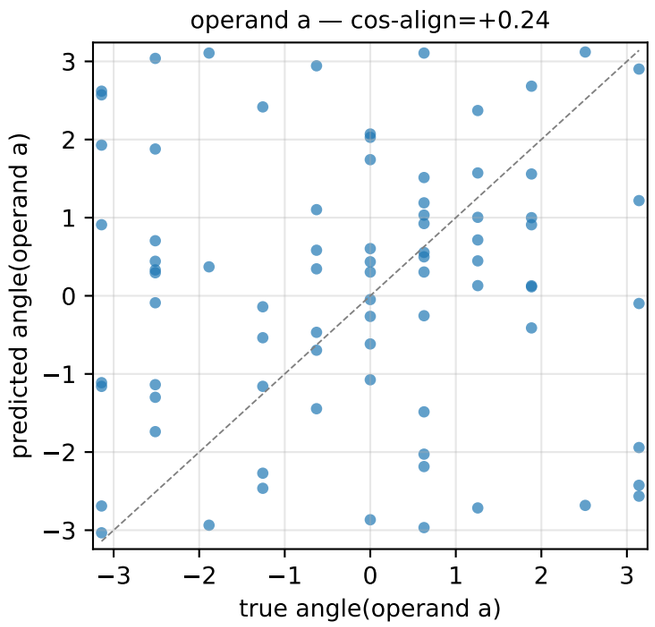


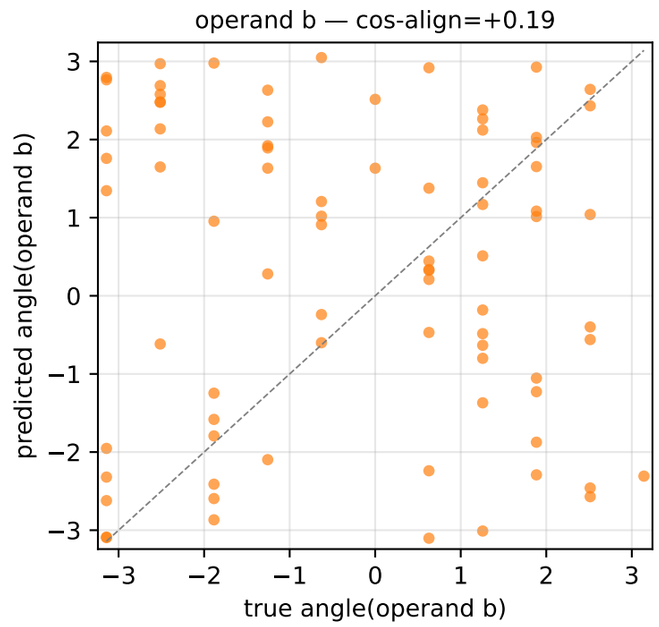


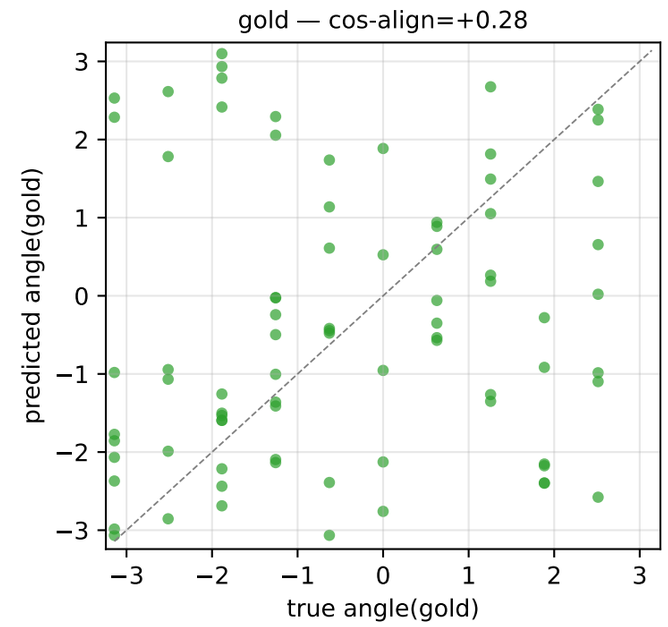


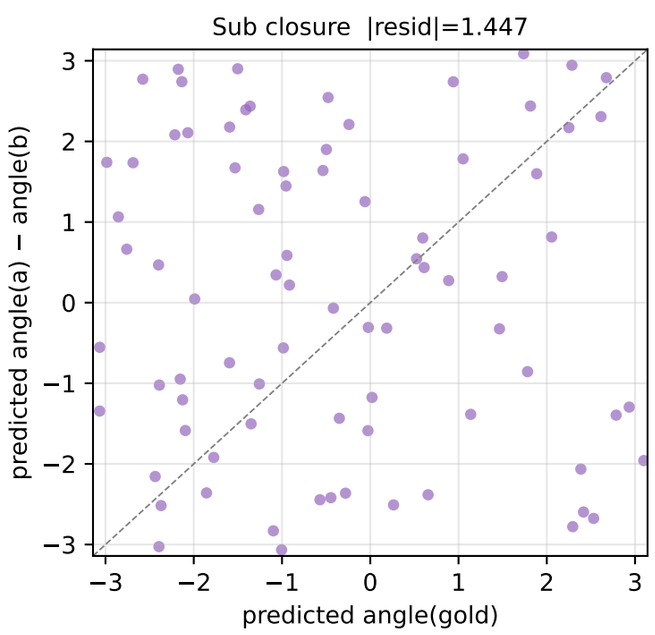

```
Period T = 5
Best cell:  step=3, layer=11
R²(a) = +0.488
cos-align(pred angle, true angle) = +0.722
max(operand)/T = 39.60
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  1.089 rad
|angle(a) + angle(b) − angle(gold)|  =  1.652 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: WEAK

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

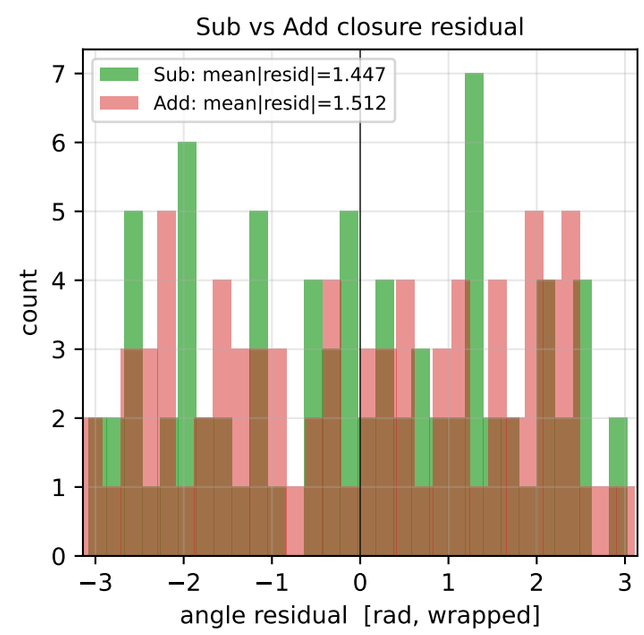
vary_numerals — Clock test at T=10





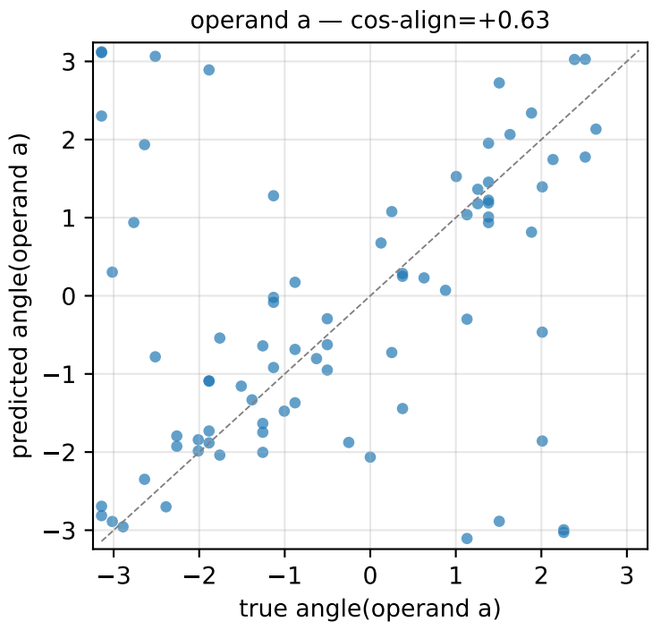


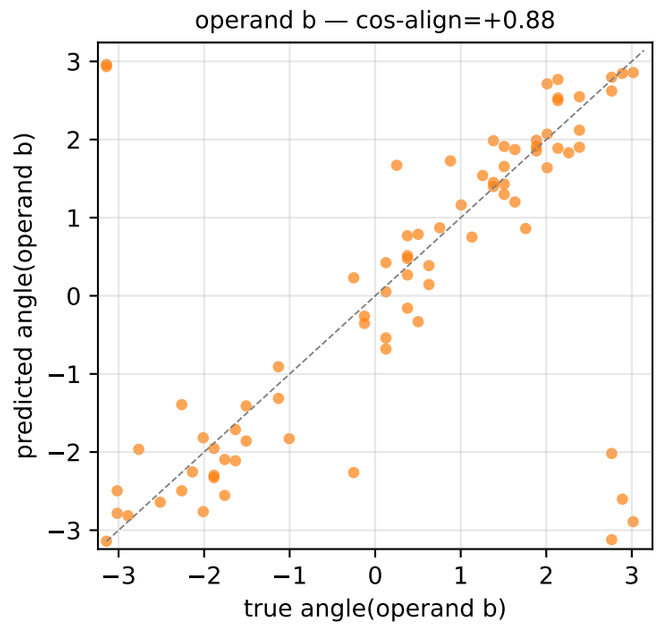


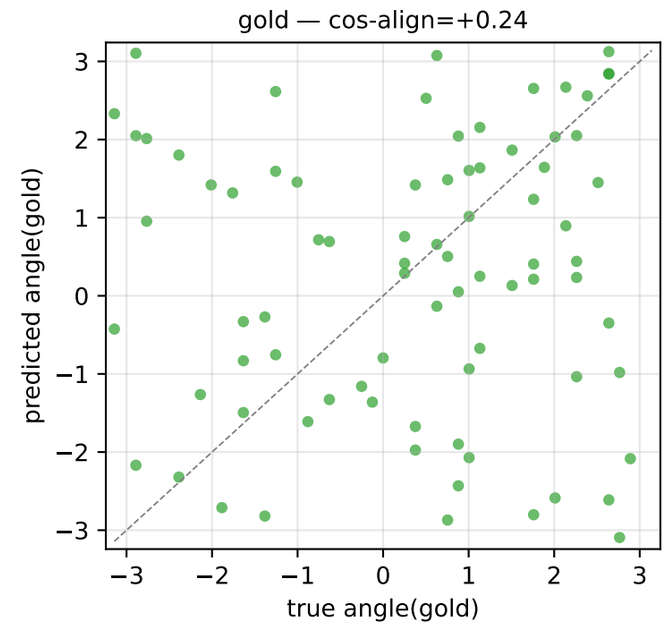


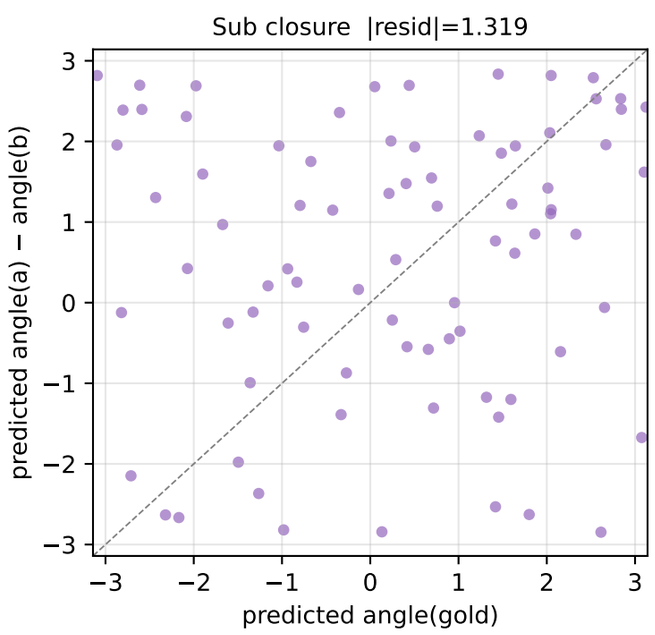

```
Period T = 10
Best cell:  step=2, layer=7
R²(a) = -0.065
cos-align(pred angle, true angle) = +0.244
max(operand)/T = 19.80
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  1.447 rad
|angle(a) + angle(b) − angle(gold)|  =  1.512 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: WEAK

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

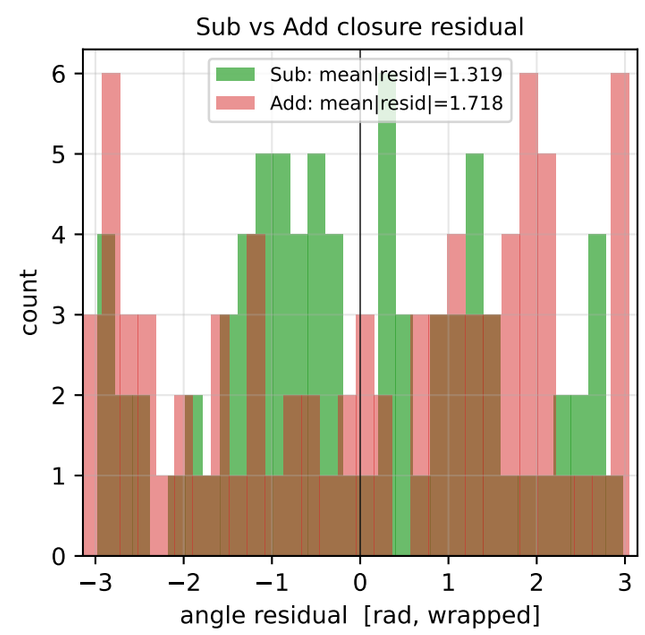
vary_numerals — Clock test at T=50





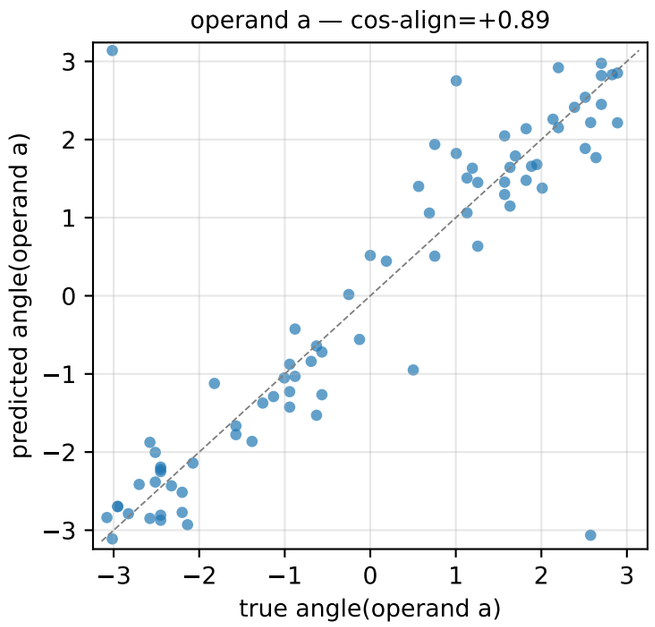


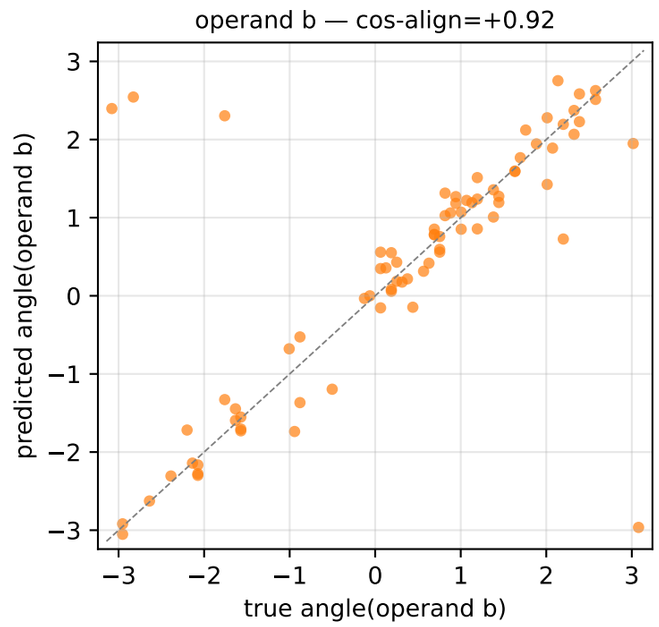


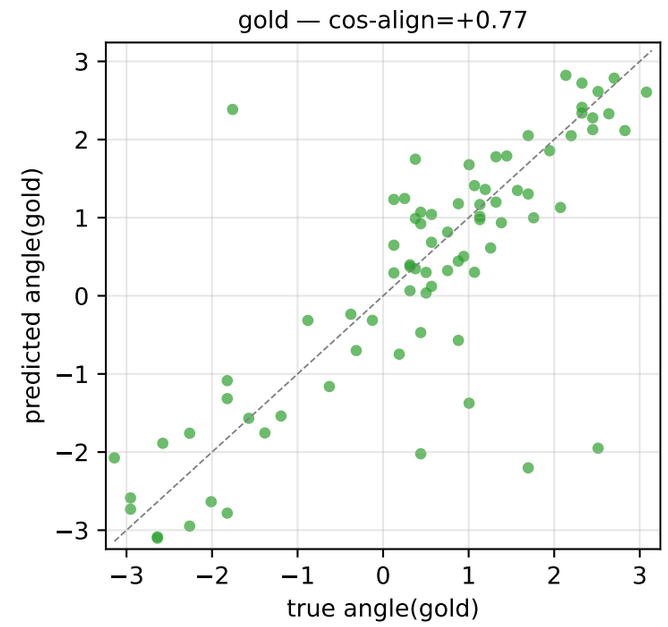


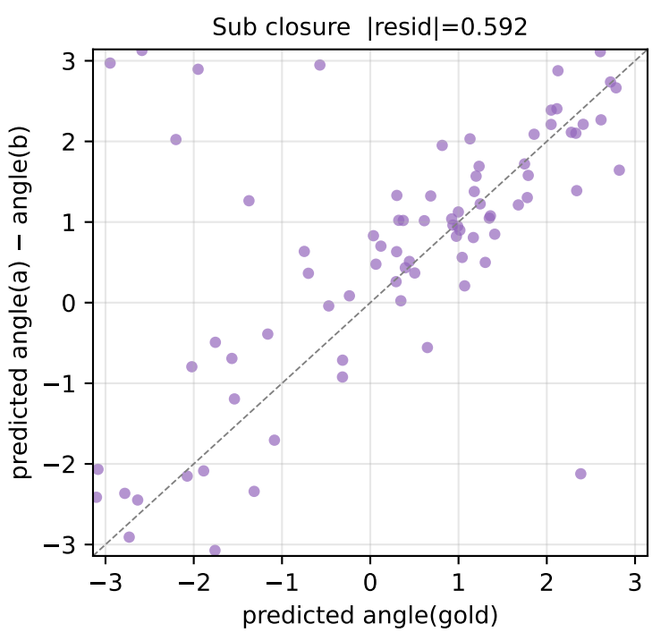

```
Period T = 50
Best cell:  step=3, layer=11
R²(a) = +0.318
cos-align(pred angle, true angle) = +0.627
max(operand)/T = 3.96
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  1.319 rad
|angle(a) + angle(b) − angle(gold)|  =  1.718 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: WEAK

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

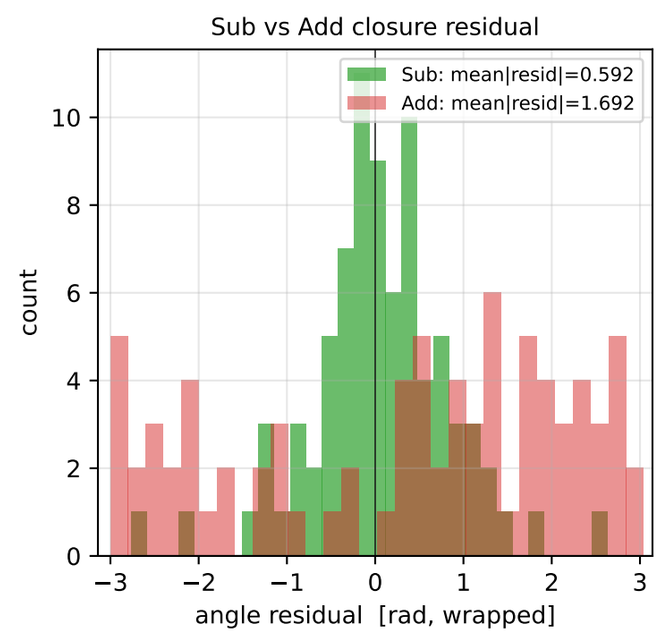
vary_numerals — Clock test at T=100





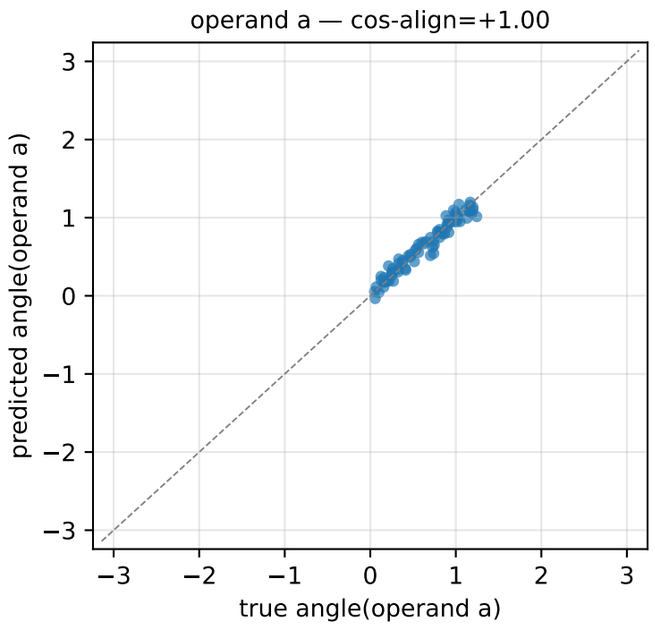


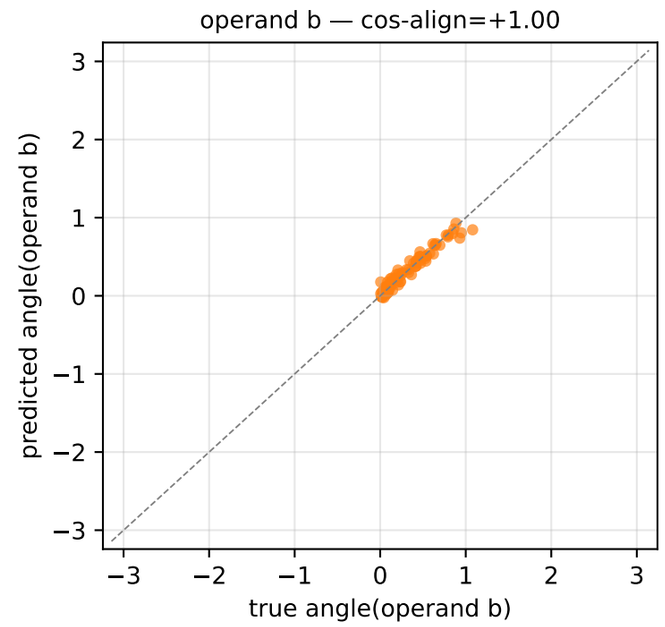


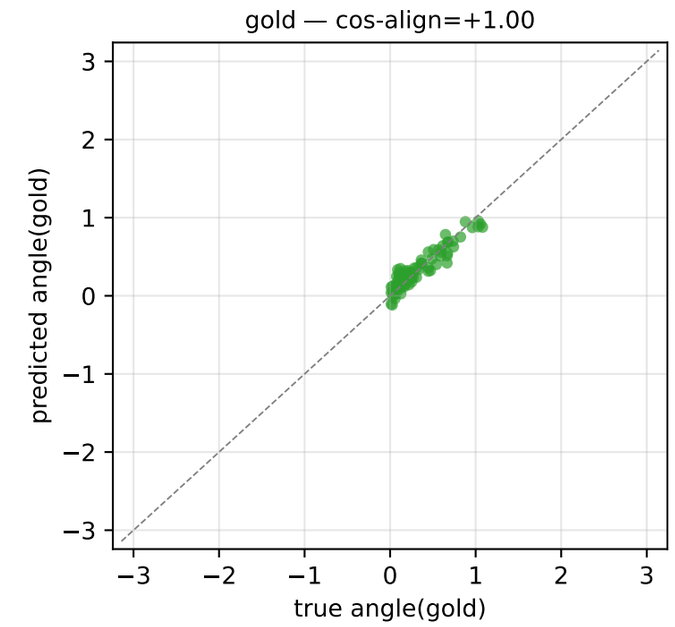


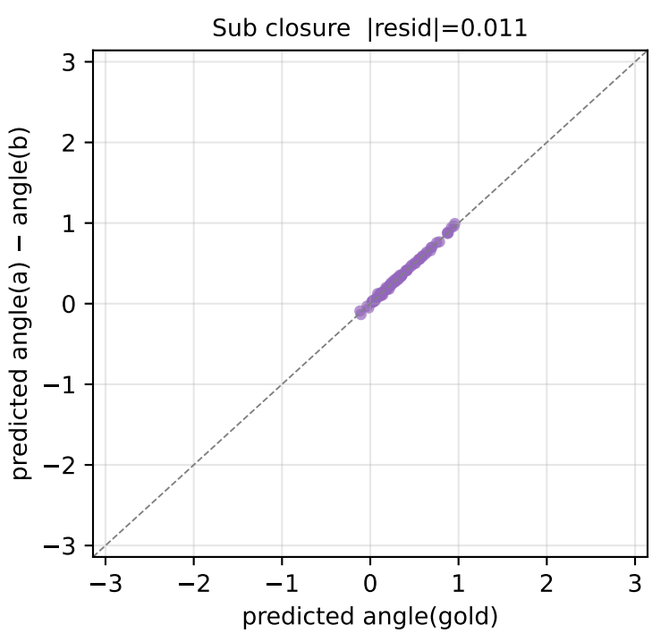

```
Period T = 100
Best cell:  step=6, layer=11
R²(a) = +0.714
cos-align(pred angle, true angle) = +0.891
max(operand)/T = 1.98
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.592 rad
|angle(a) + angle(b) − angle(gold)|  =  1.692 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

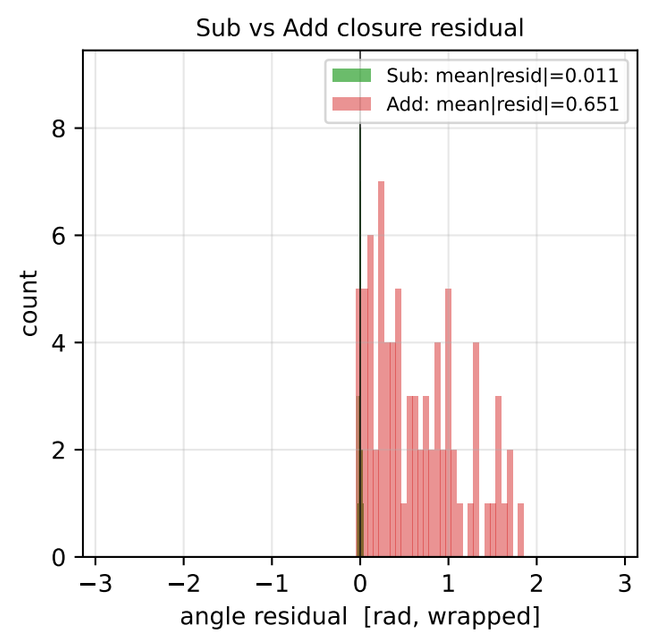
vary_numerals — Clock test at T=1000





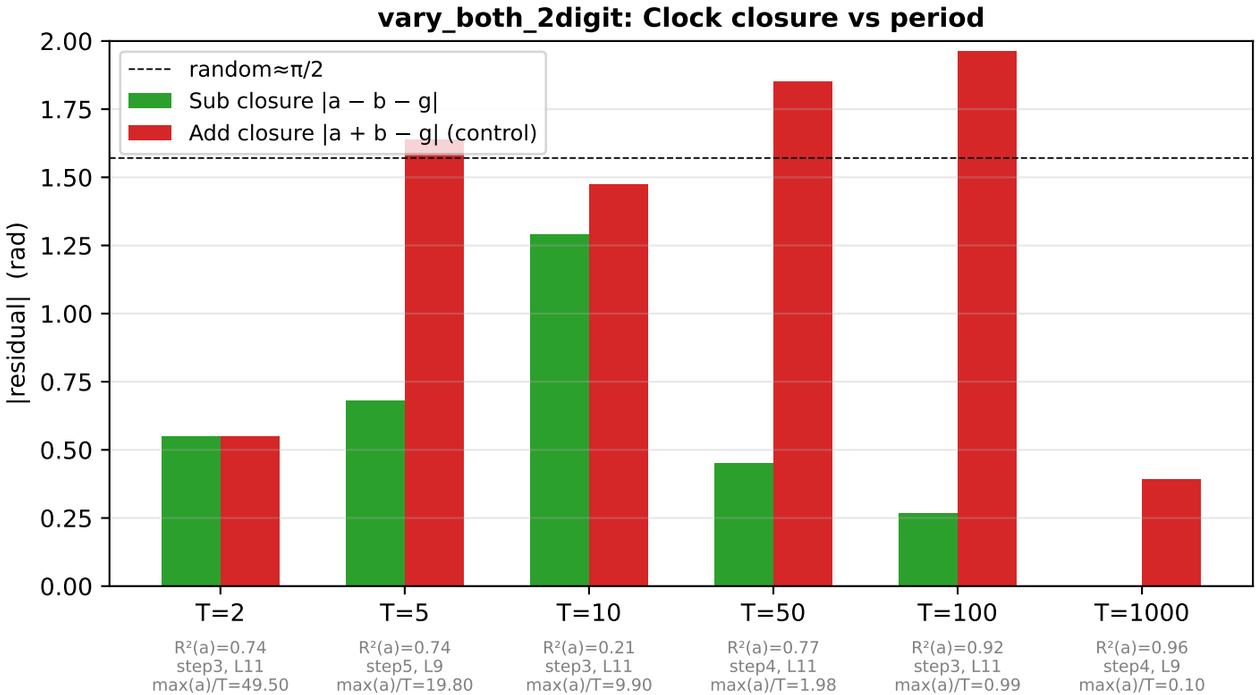


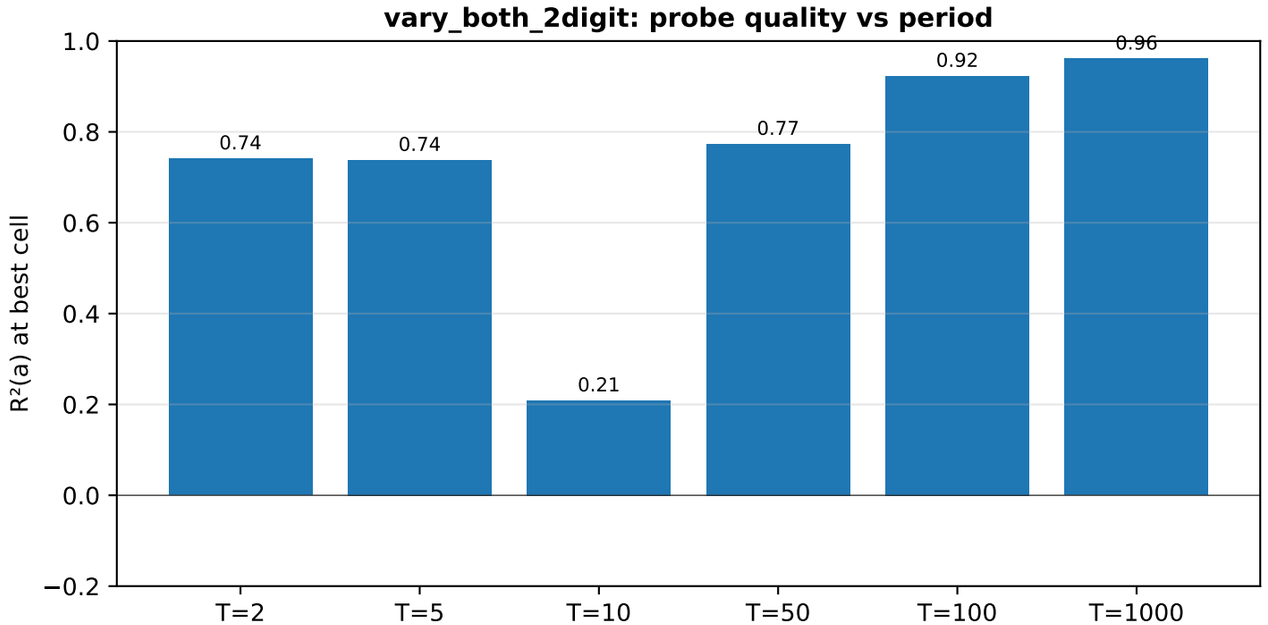




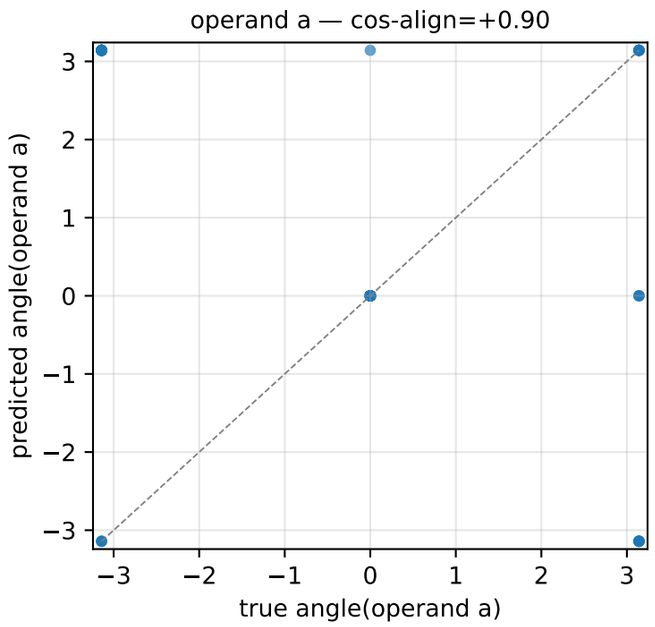

```
Period T = 1000
Best cell:  step=4, layer=9
R²(a) = +0.941
cos-align(pred angle, true angle) = +0.997
max(operand)/T = 0.20
  Non-trivial Clock test? NO (near-linear)

|angle(a) − angle(b) − angle(gold)|  =  0.011 rad
|angle(a) + angle(b) − angle(gold)|  =  0.651 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

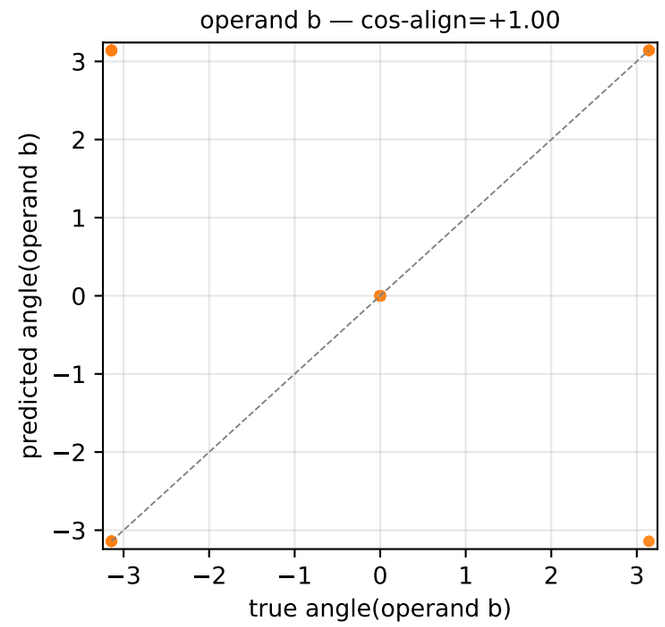
Clock test on CODI-GPT-2 latent residuals: vary_both_2digit

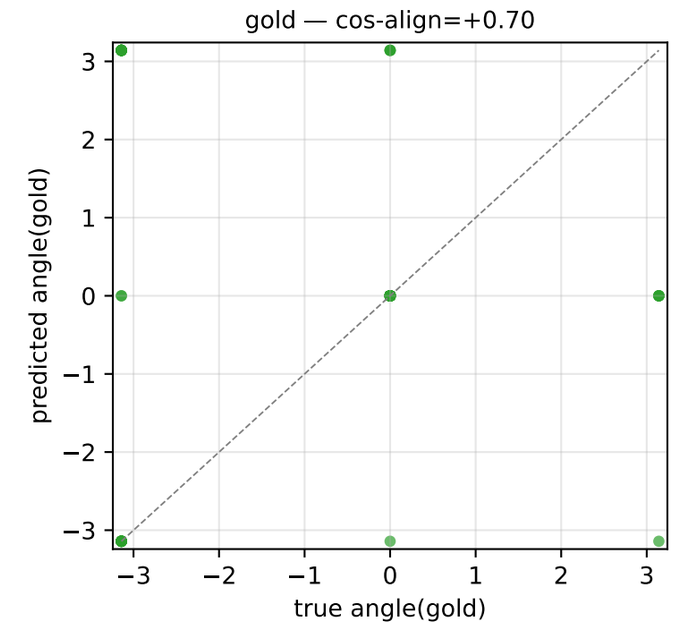


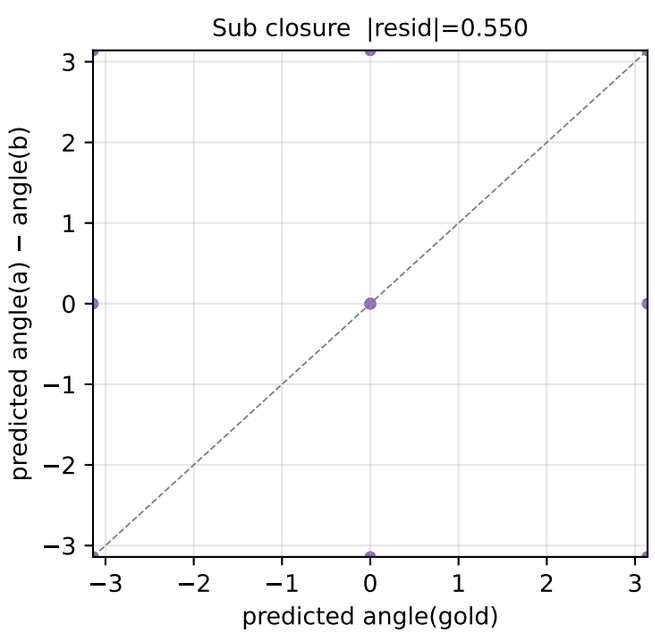


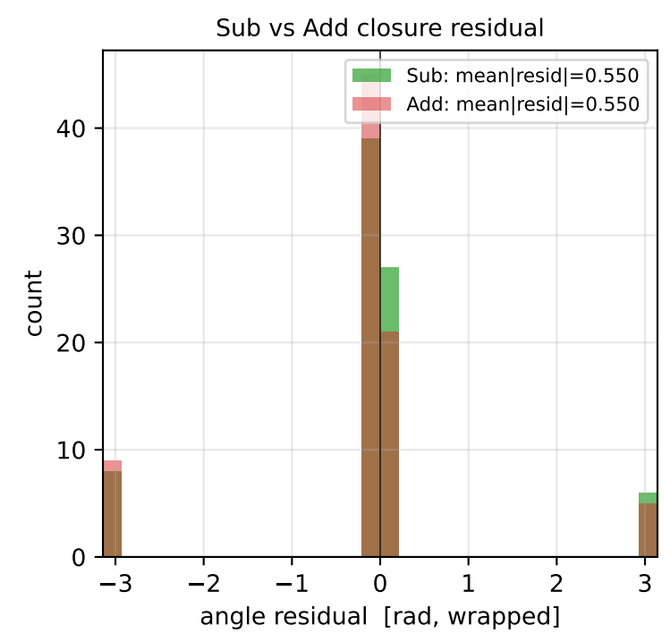
vary_both_2digit — Clock test at T=2





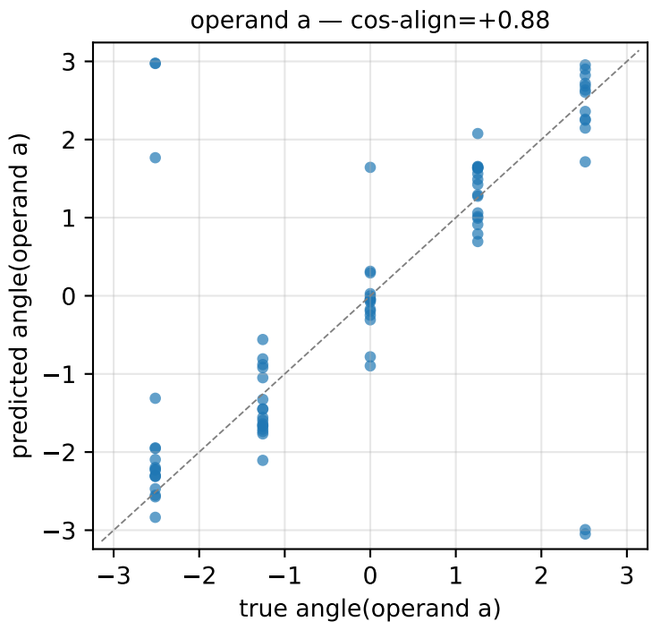


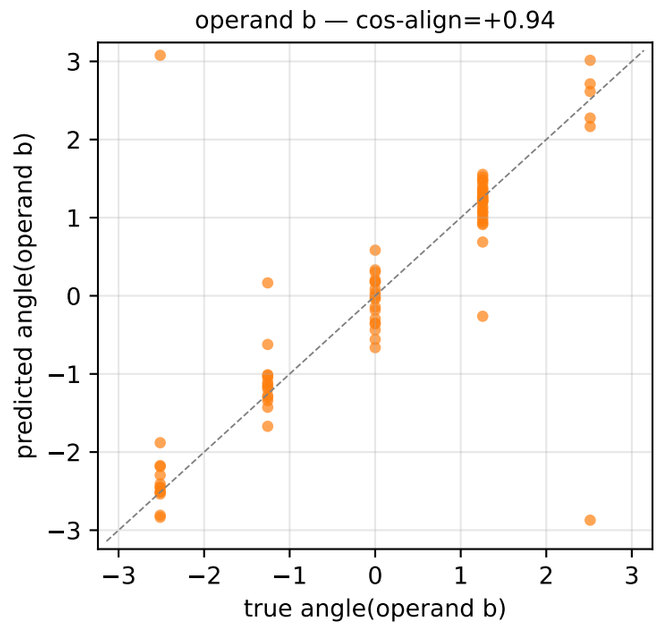


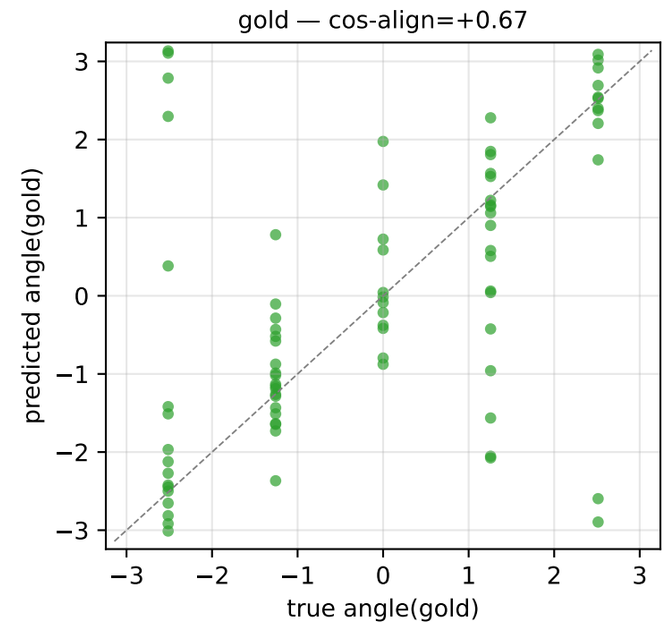


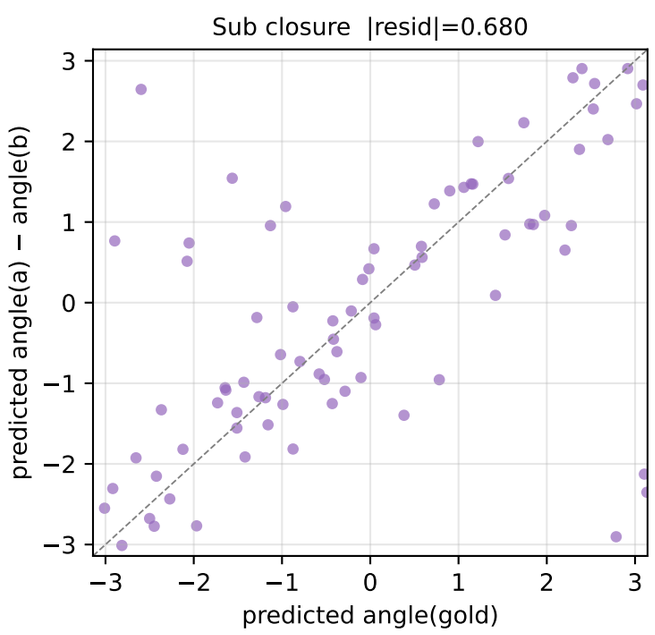

```
Period T = 2
Best cell:  step=3, layer=11
R²(a) = +0.742
cos-align(pred angle, true angle) = +0.900
max(operand)/T = 49.50
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.550 rad
|angle(a) + angle(b) − angle(gold)|  =  0.550 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

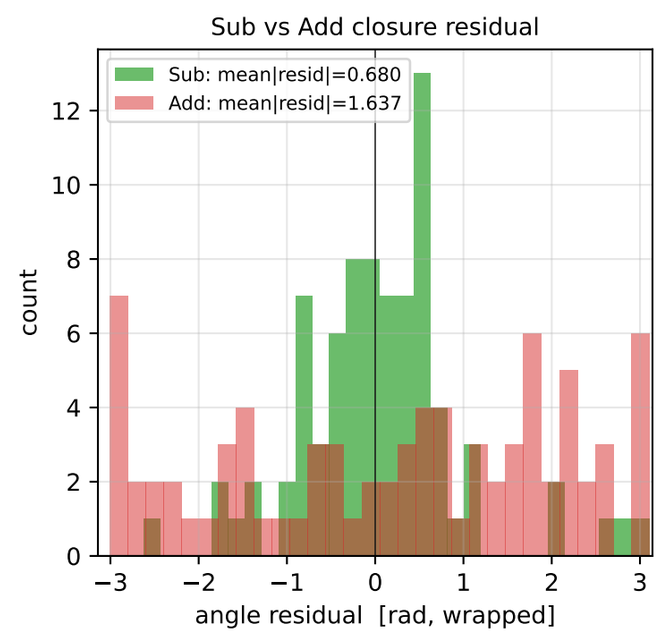
vary_both_2digit — Clock test at T=5





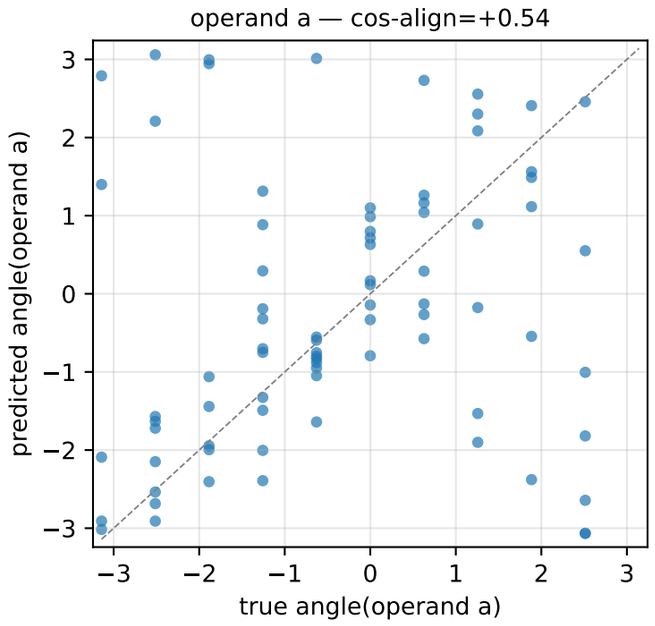


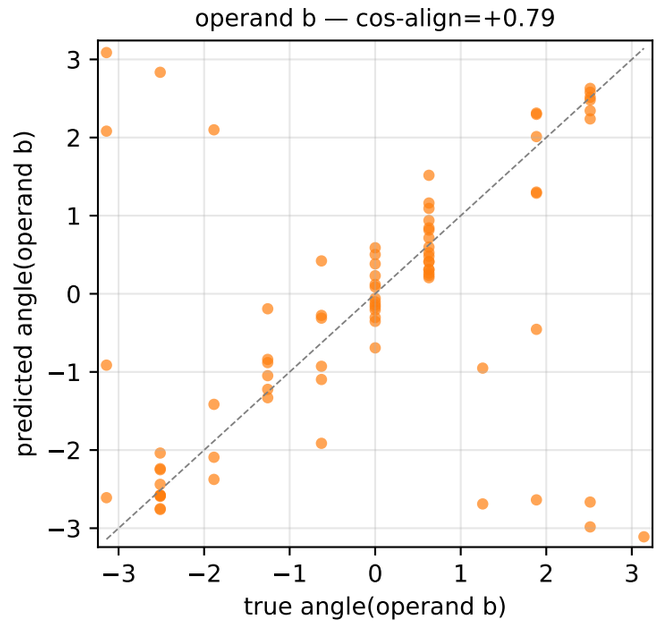


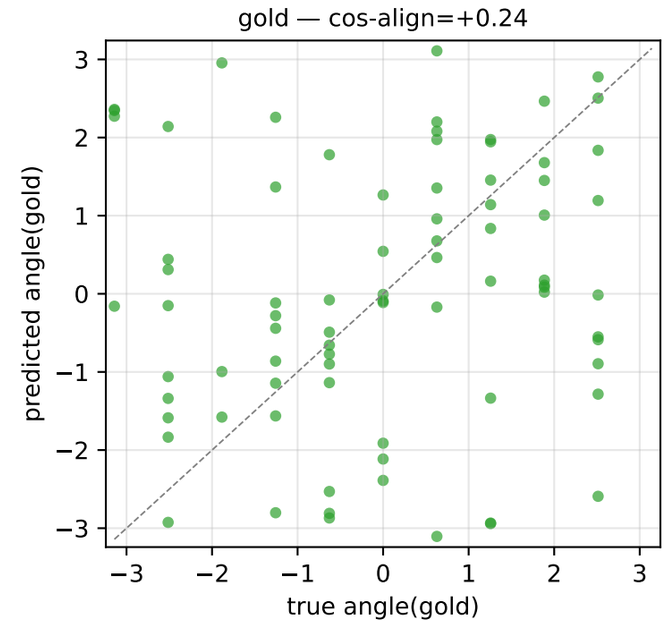


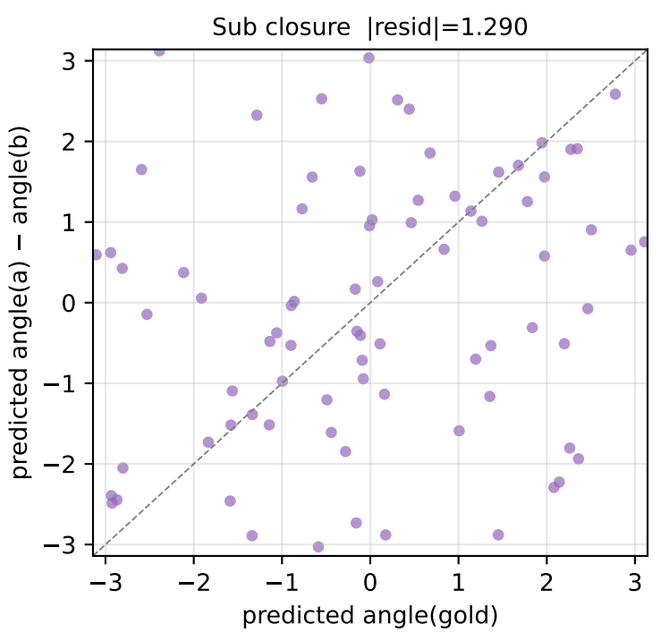

```
Period T = 5
Best cell:  step=5, layer=9
R²(a) = +0.738
cos-align(pred angle, true angle) = +0.885
max(operand)/T = 19.80
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.680 rad
|angle(a) + angle(b) − angle(gold)|  =  1.637 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

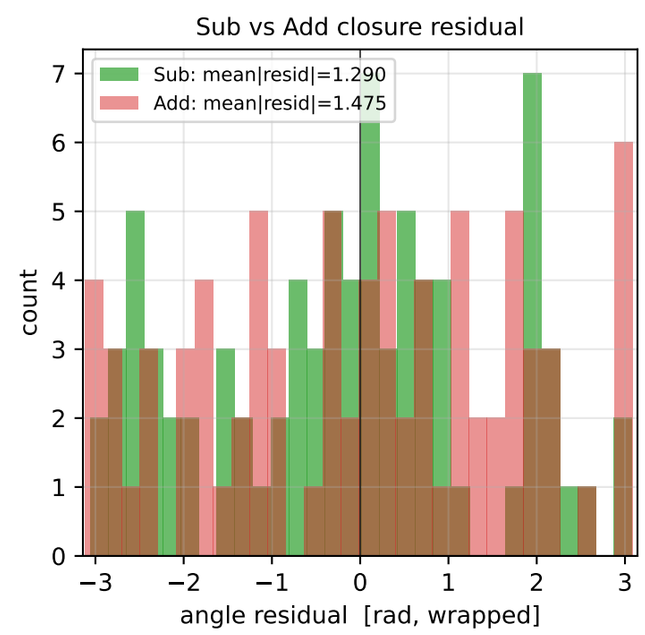
vary_both_2digit — Clock test at T=10





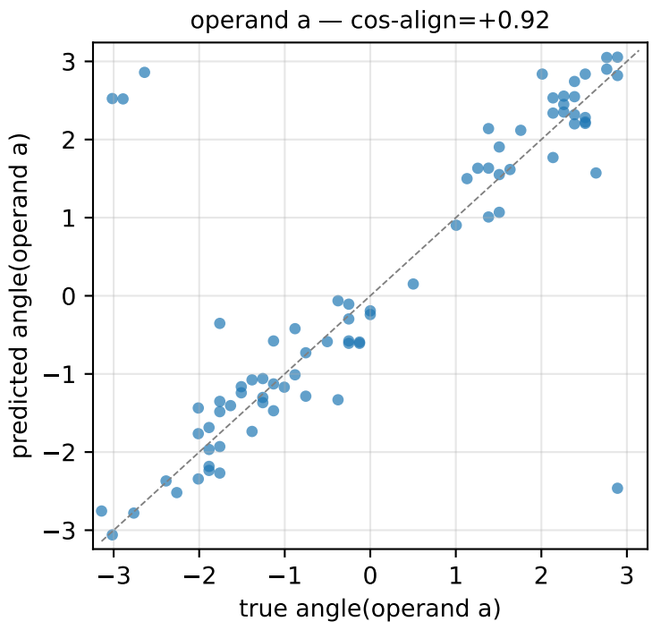


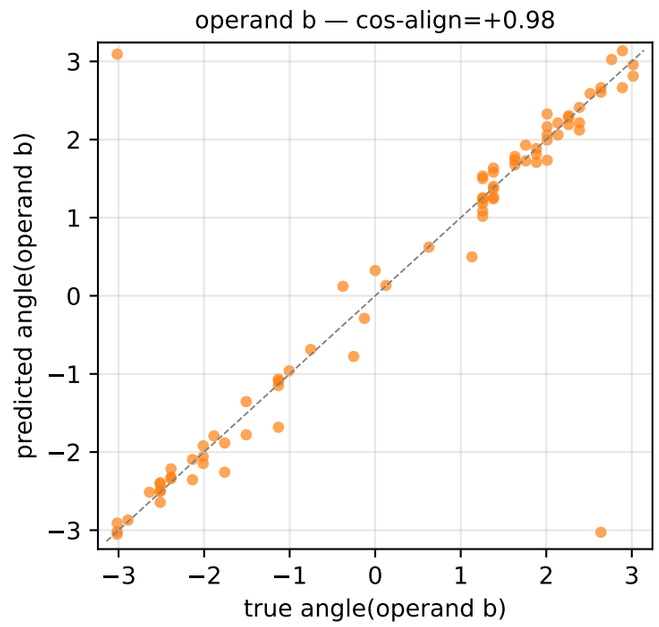


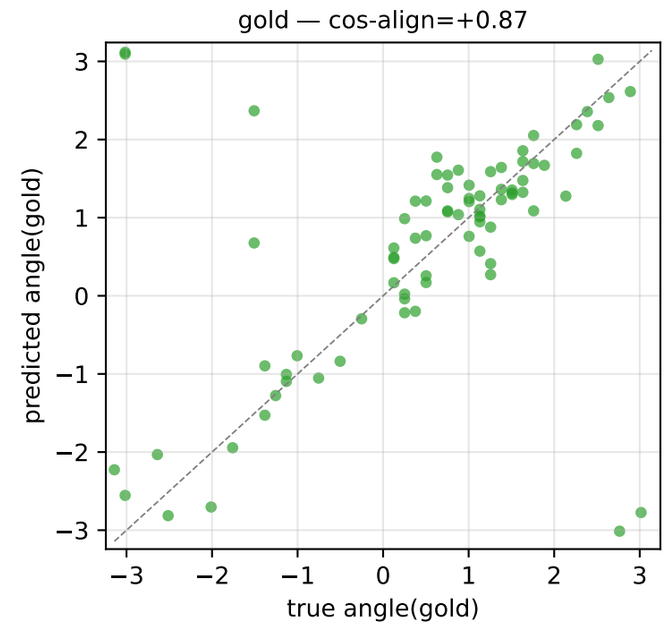


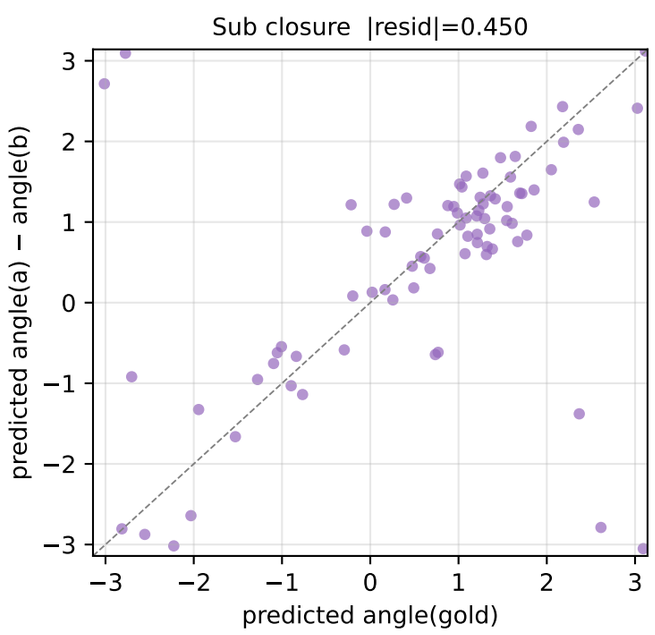

```
Period T = 10
Best cell:  step=3, layer=11
R²(a) = +0.209
cos-align(pred angle, true angle) = +0.540
max(operand)/T = 9.90
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  1.290 rad
|angle(a) + angle(b) − angle(gold)|  =  1.475 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: WEAK

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

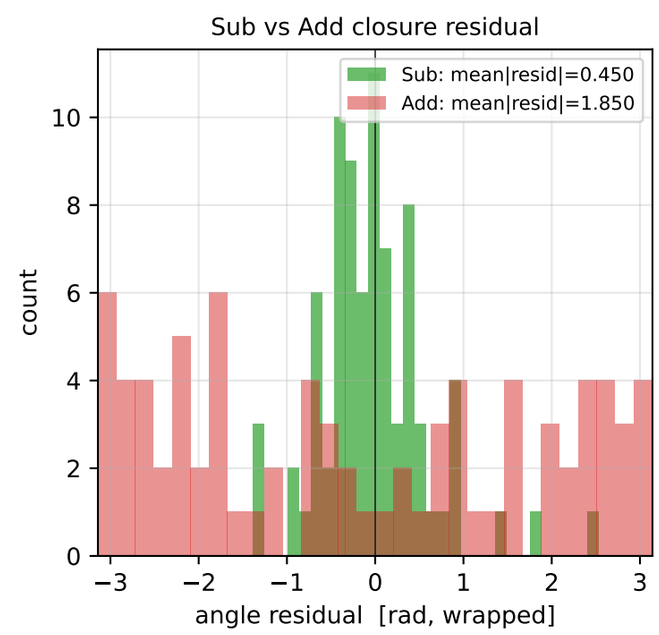
vary_both_2digit — Clock test at T=50





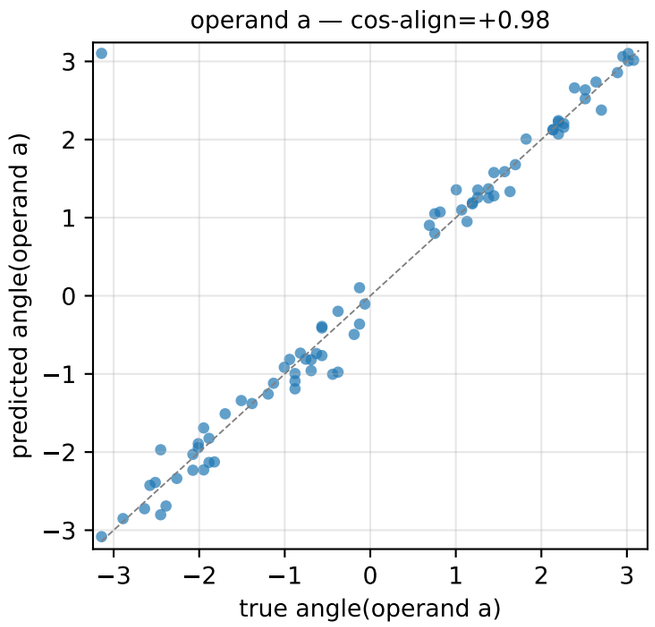


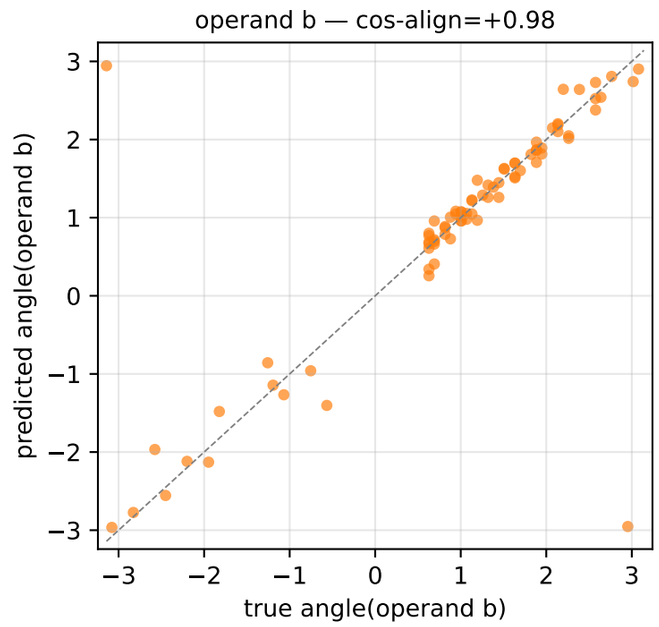


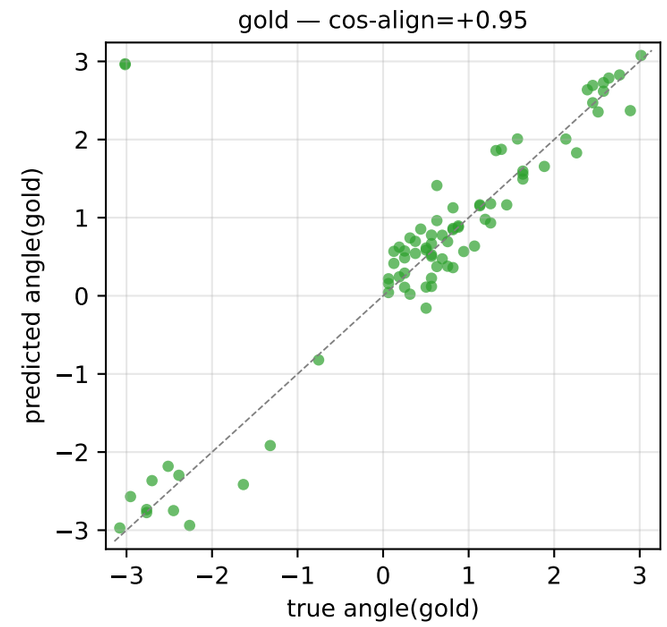


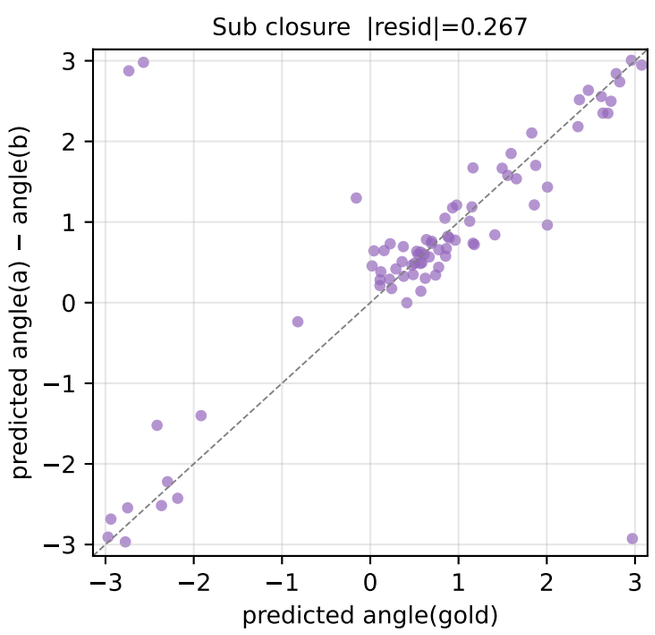

```
Period T = 50
Best cell:  step=4, layer=11
R²(a) = +0.773
cos-align(pred angle, true angle) = +0.916
max(operand)/T = 1.98
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.450 rad
|angle(a) + angle(b) − angle(gold)|  =  1.850 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

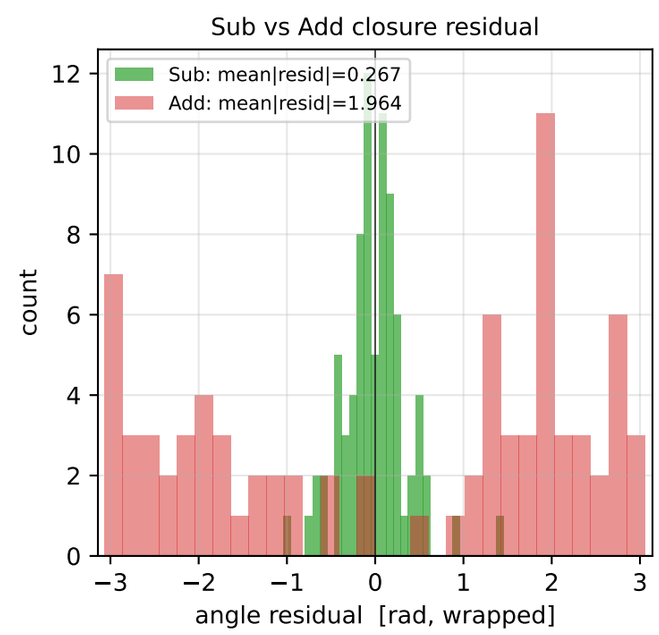
vary_both_2digit — Clock test at T=100





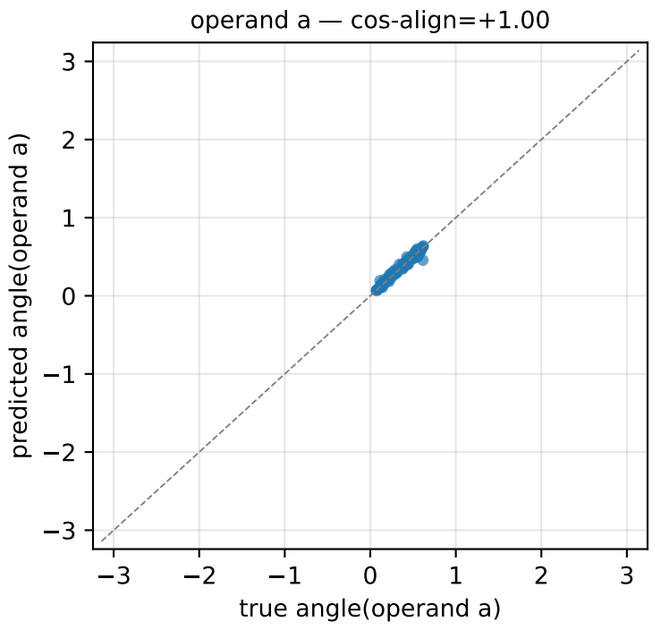


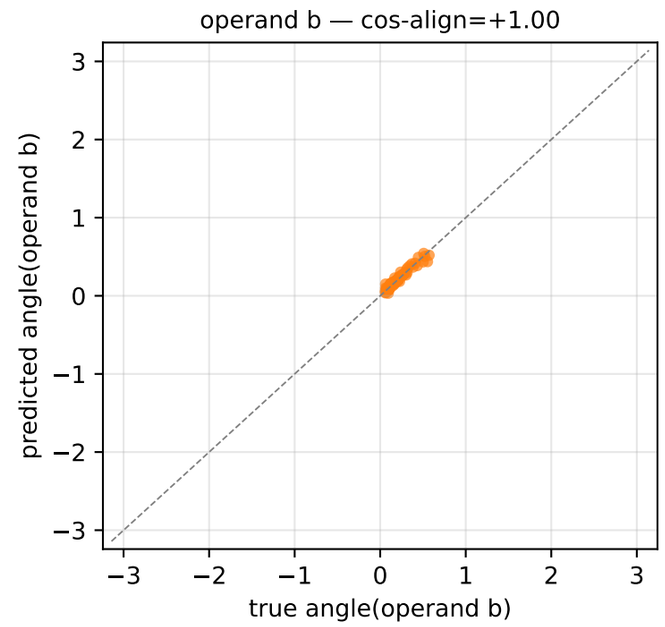


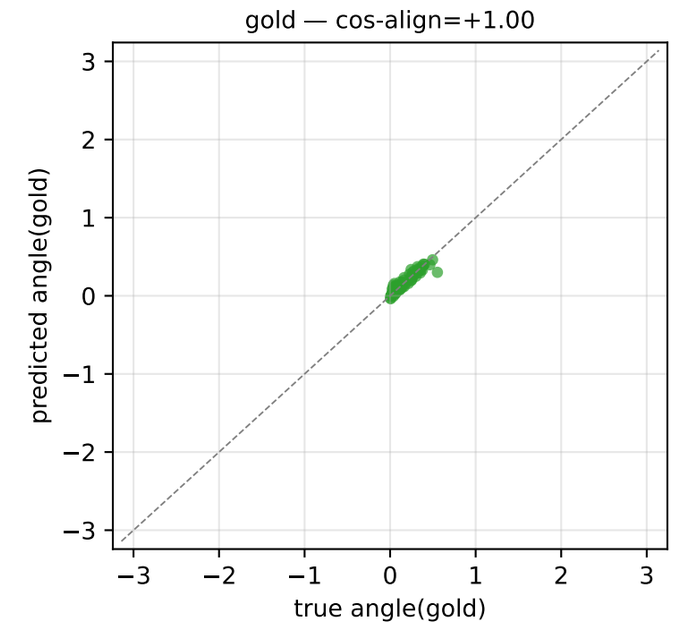


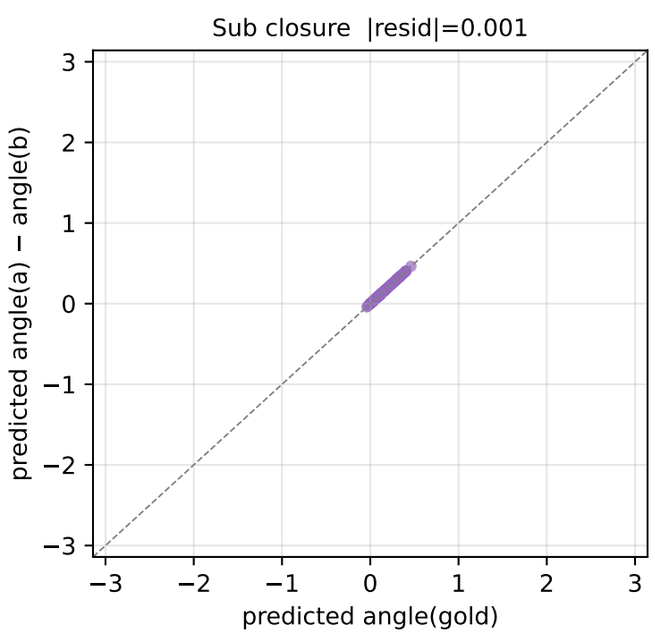

```
Period T = 100
Best cell:  step=3, layer=11
R²(a) = +0.923
cos-align(pred angle, true angle) = +0.981
max(operand)/T = 0.99
  Non-trivial Clock test? NO (near-linear)

|angle(a) − angle(b) − angle(gold)|  =  0.267 rad
|angle(a) + angle(b) − angle(gold)|  =  1.964 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

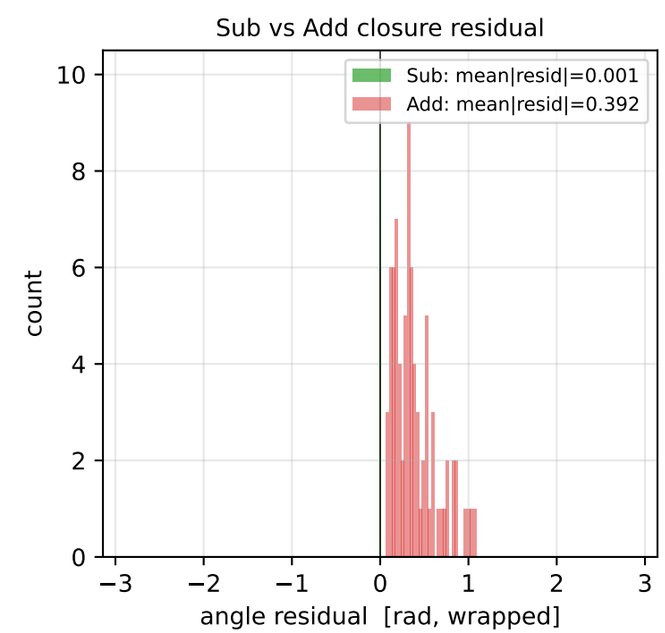
vary_both_2digit — Clock test at T=1000





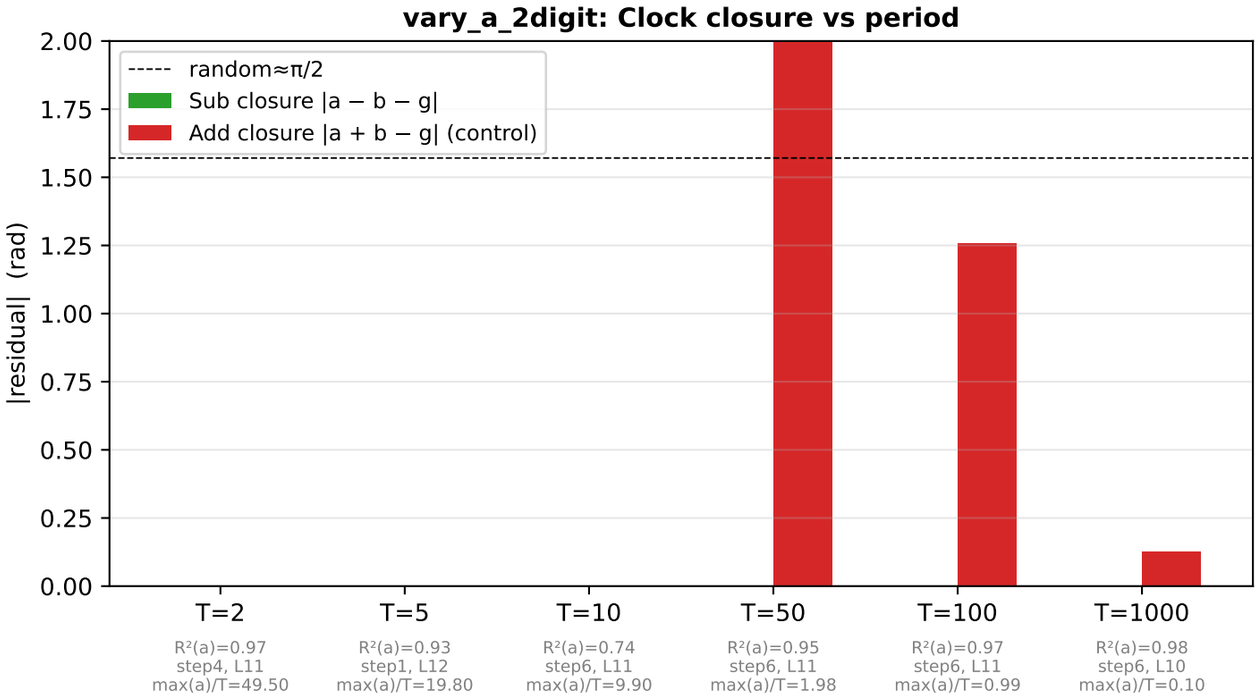


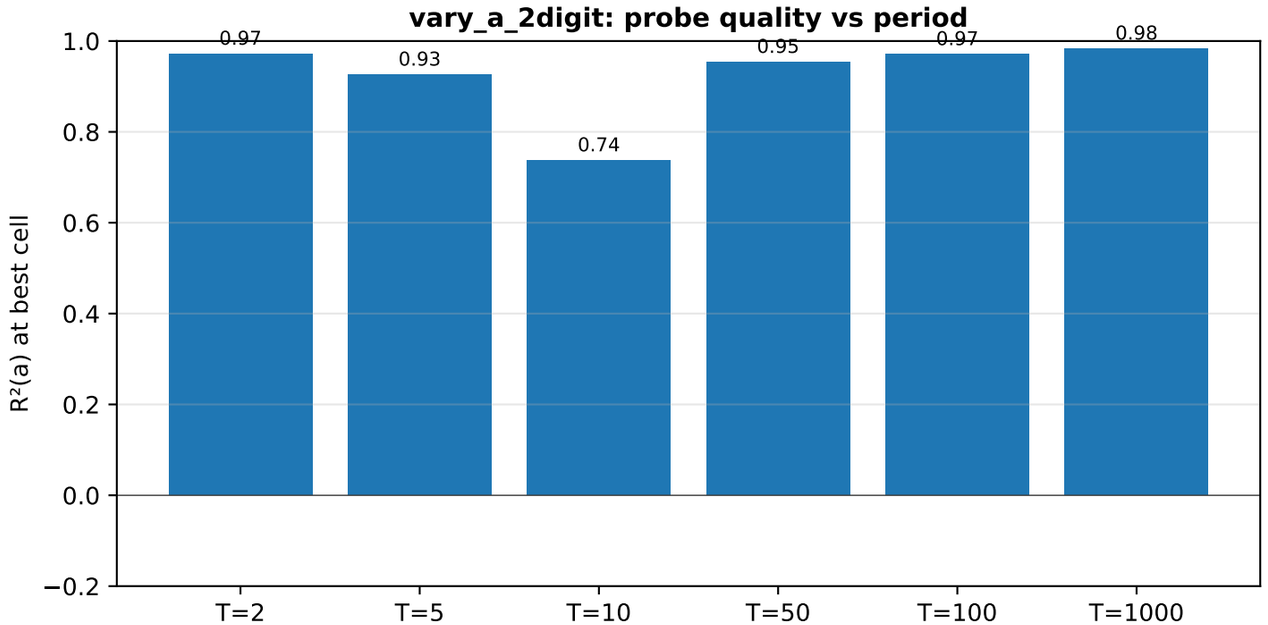




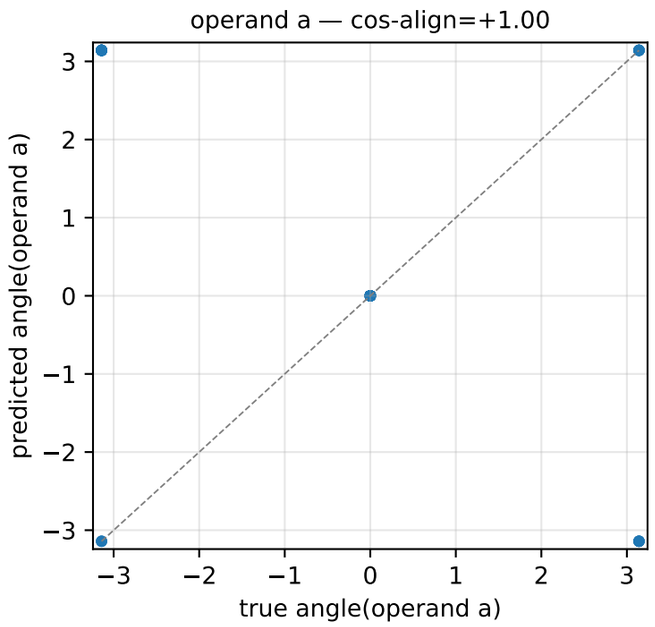

```
Period T = 1000
Best cell:  step=4, layer=9
R²(a) = +0.961
cos-align(pred angle, true angle) = +0.999
max(operand)/T = 0.10
  Non-trivial Clock test? NO (near-linear)

|angle(a) − angle(b) − angle(gold)|  =  0.001 rad
|angle(a) + angle(b) − angle(gold)|  =  0.392 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

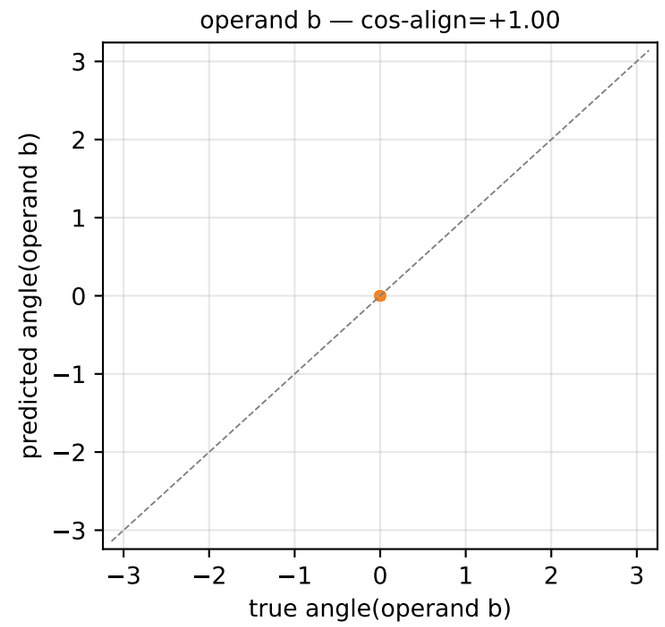
Clock test on CODI-GPT-2 latent residuals: vary_a_2digit

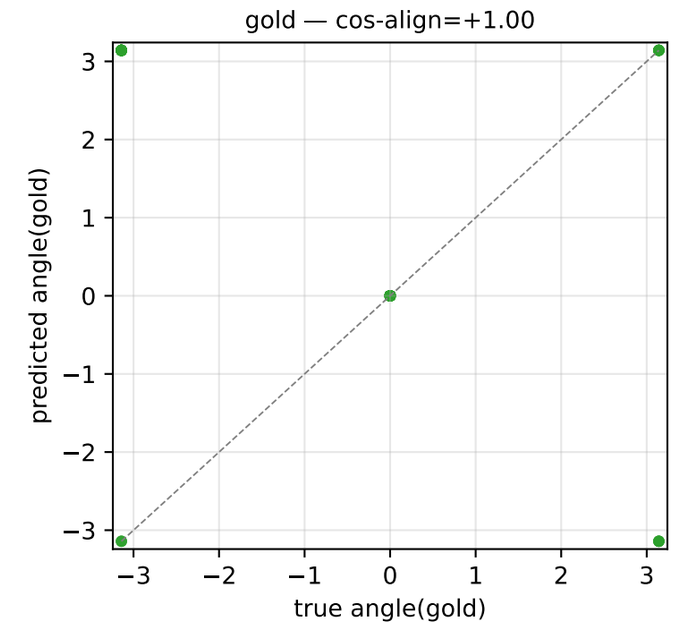


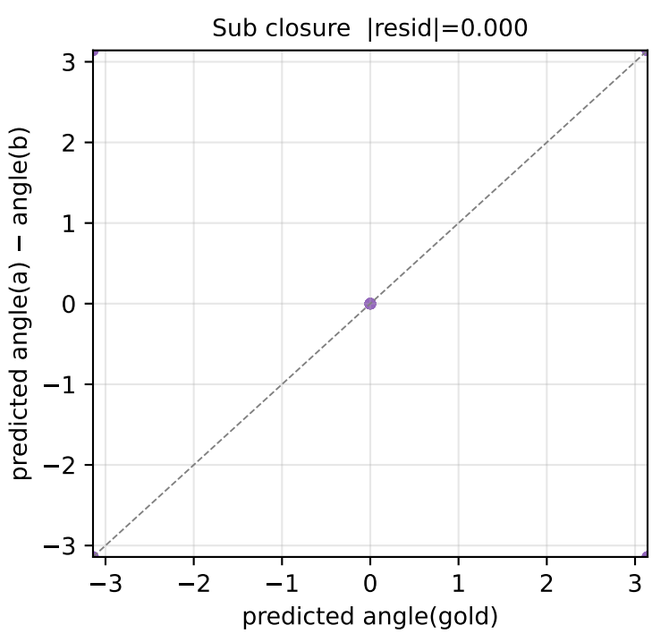


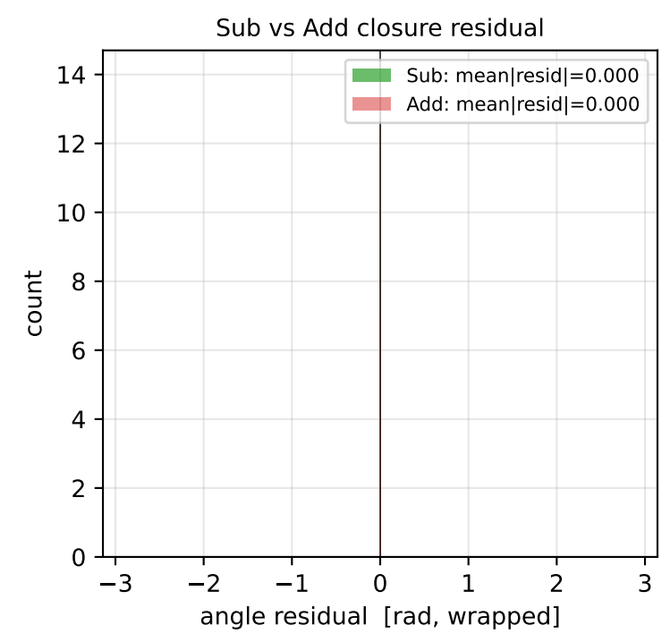
vary_a_2digit — Clock test at T=2





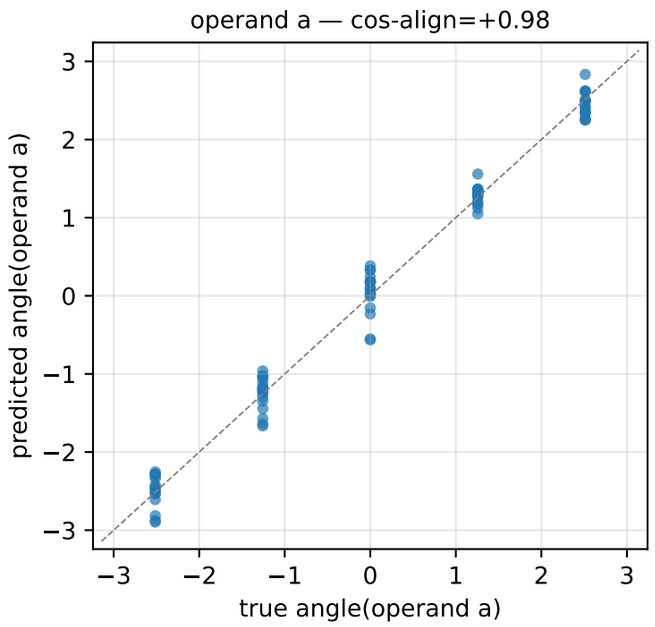


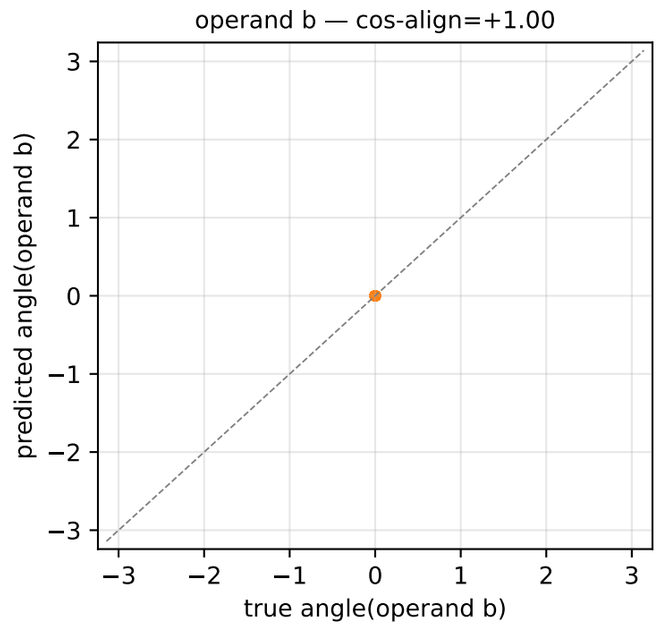


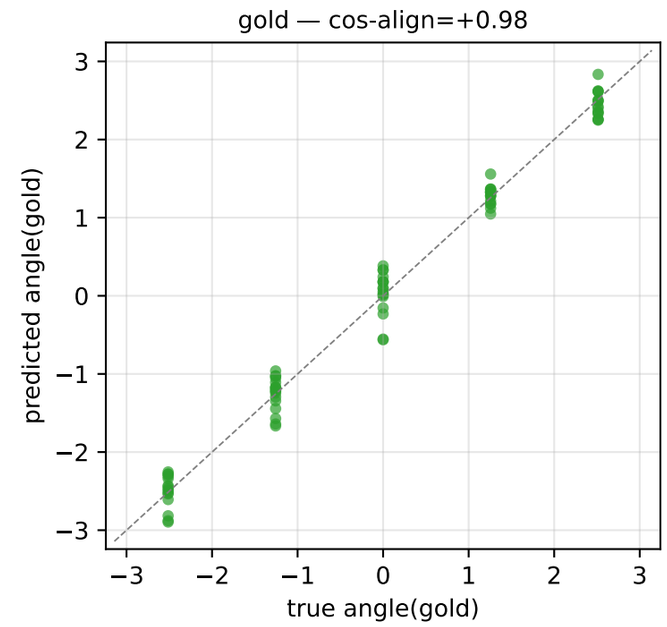


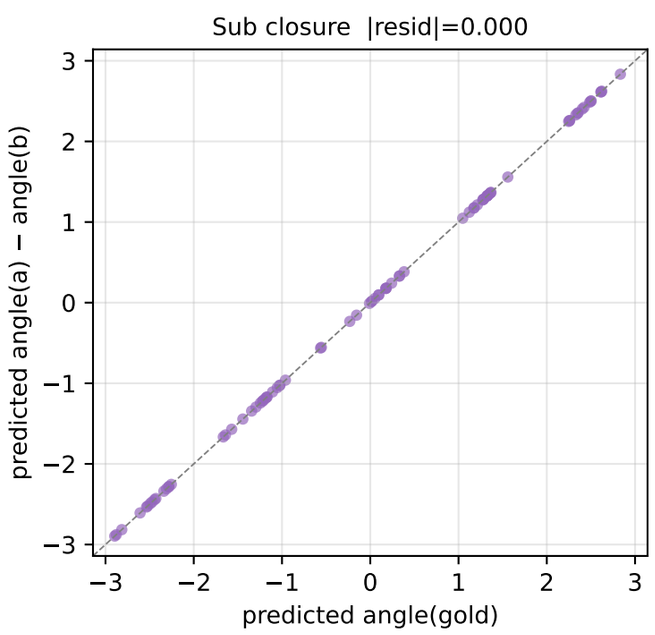

```
Period T = 2
Best cell:  step=4, layer=11
R²(a) = +0.972
cos-align(pred angle, true angle) = +1.000
max(operand)/T = 49.50
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  0.000 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

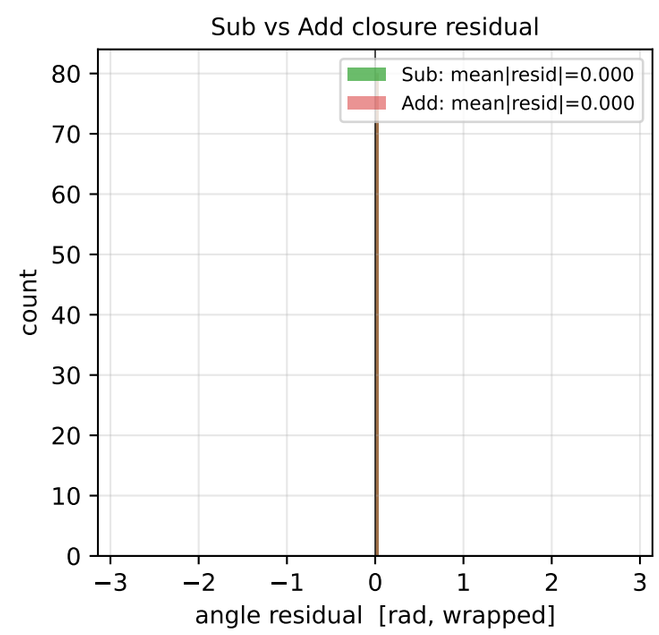
vary_a_2digit — Clock test at T=5





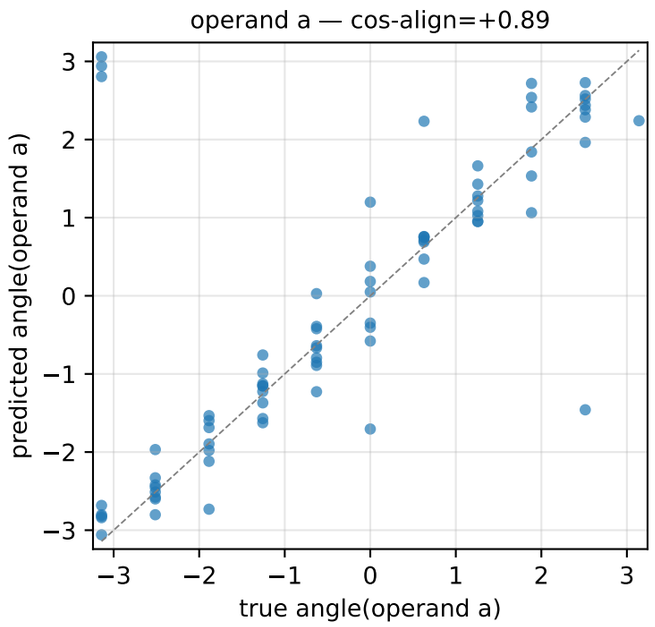


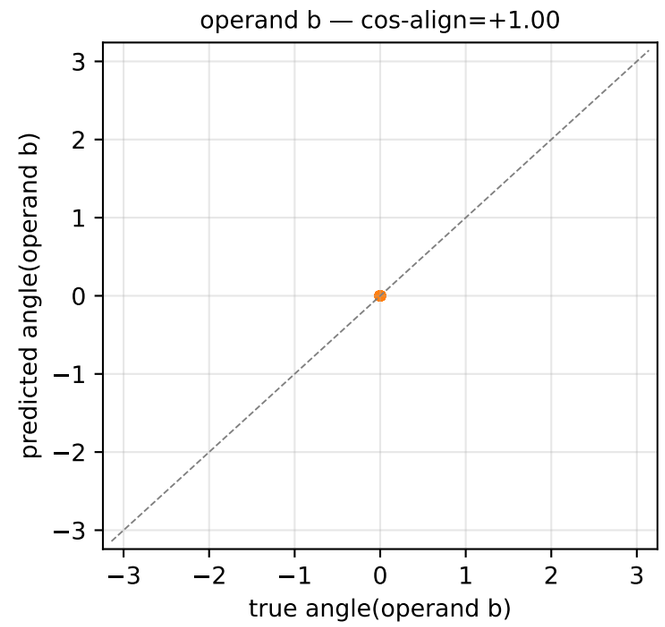


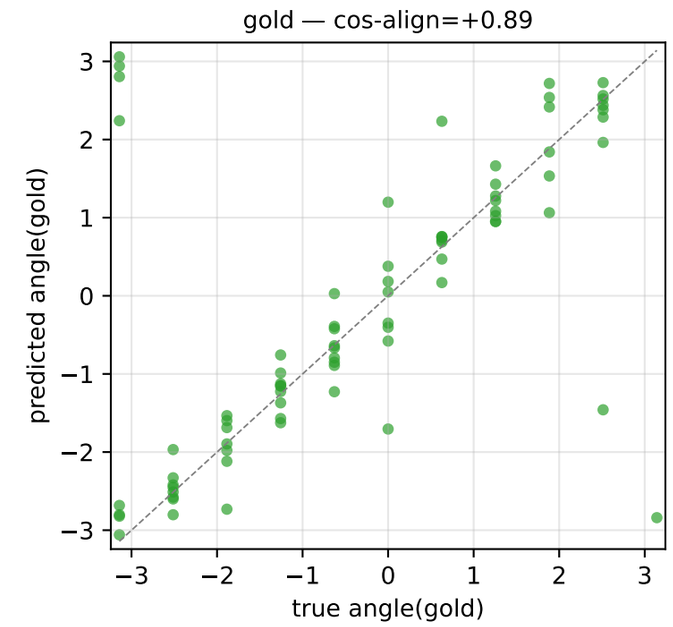


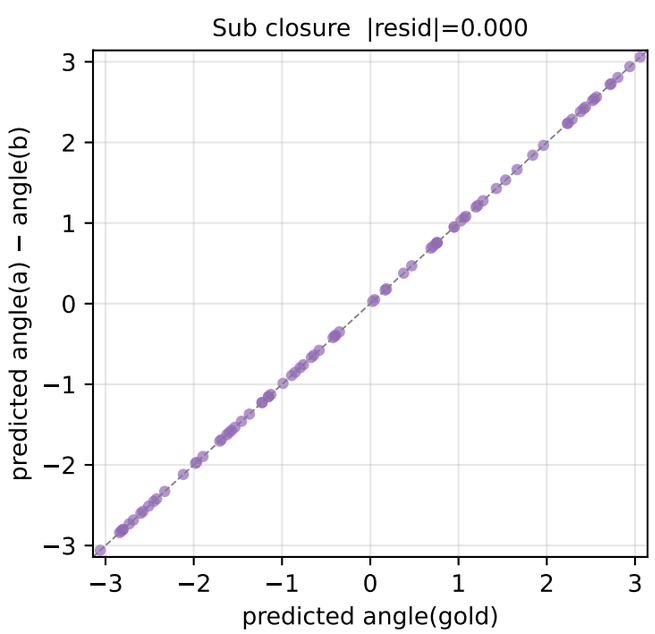

```
Period T = 5
Best cell:  step=1, layer=12
R²(a) = +0.926
cos-align(pred angle, true angle) = +0.980
max(operand)/T = 19.80
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  0.000 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

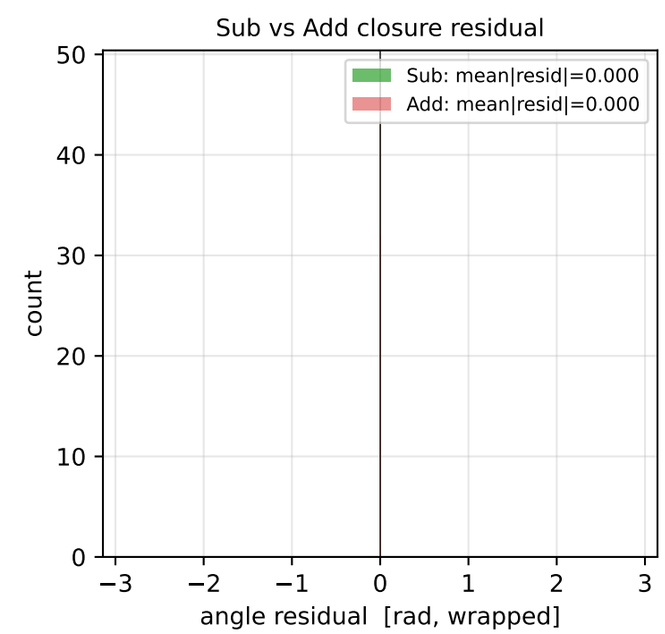
vary_a_2digit — Clock test at T=10





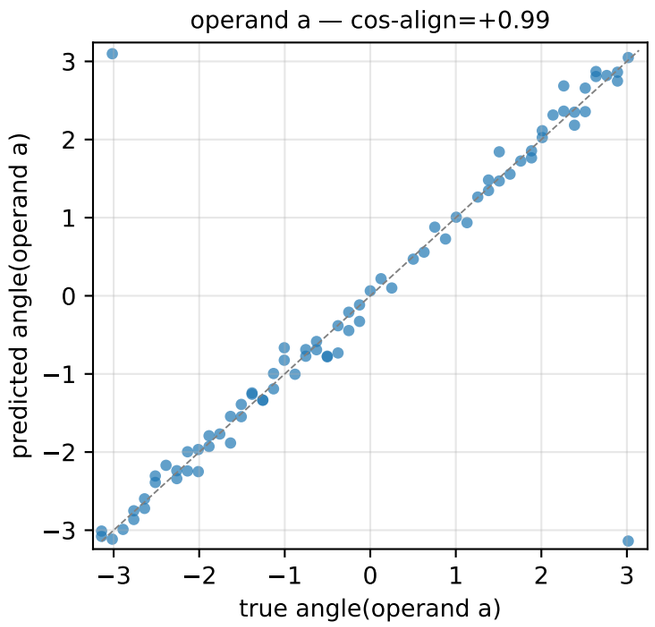


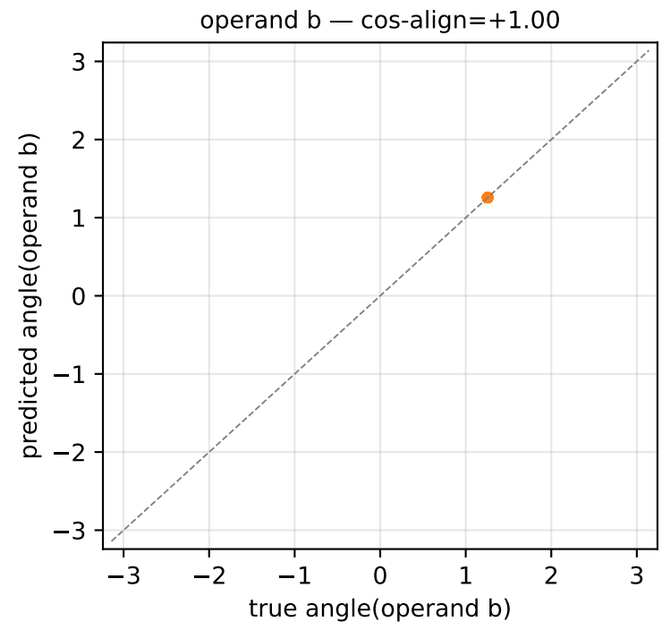


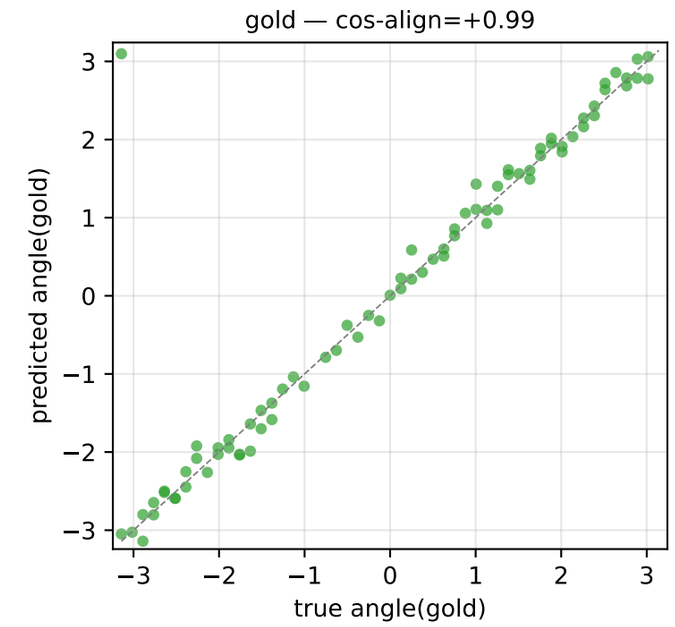


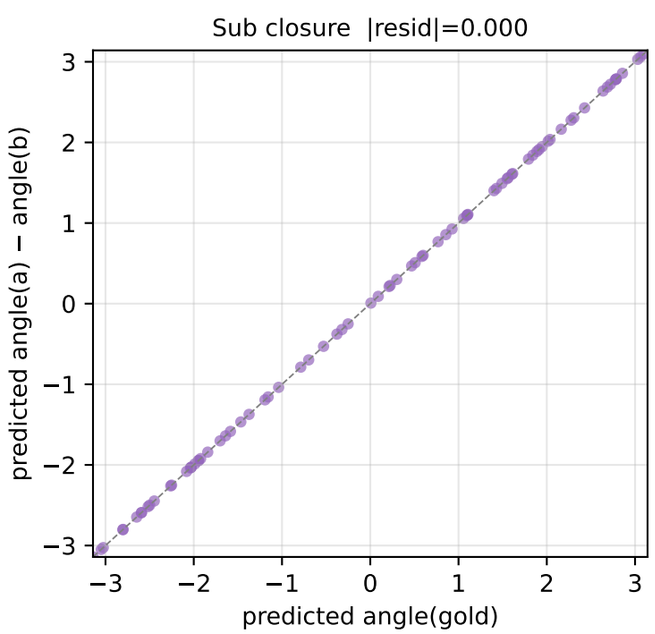

```
Period T = 10
Best cell:  step=6, layer=11
R²(a) = +0.737
cos-align(pred angle, true angle) = +0.891
max(operand)/T = 9.90
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  0.000 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

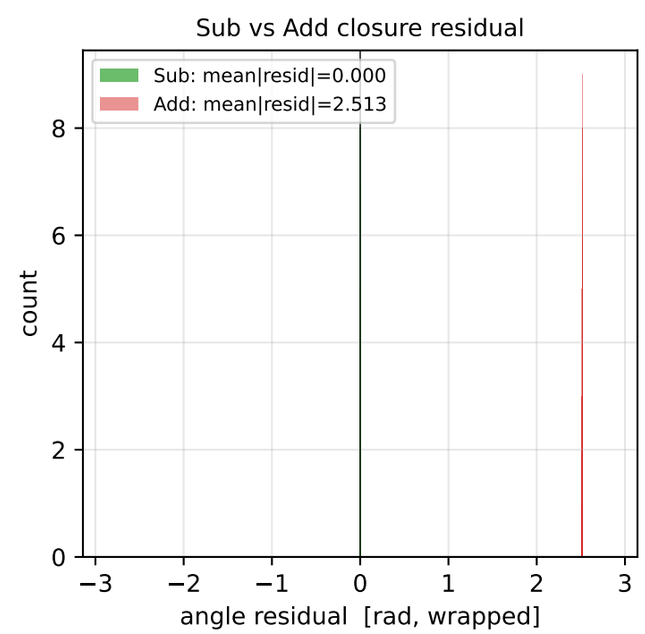
vary_a_2digit — Clock test at T=50





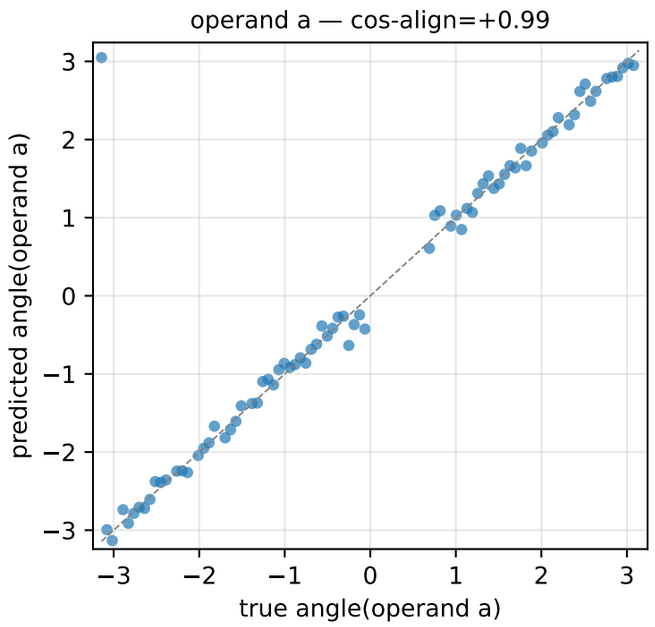


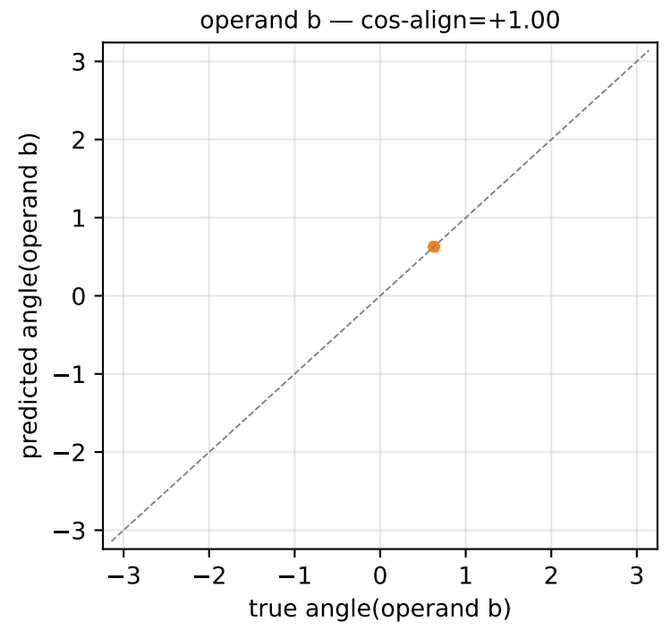


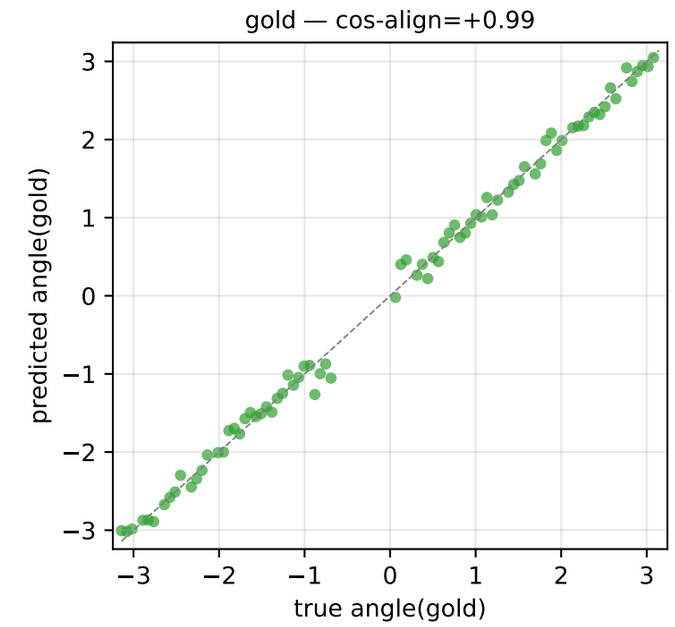


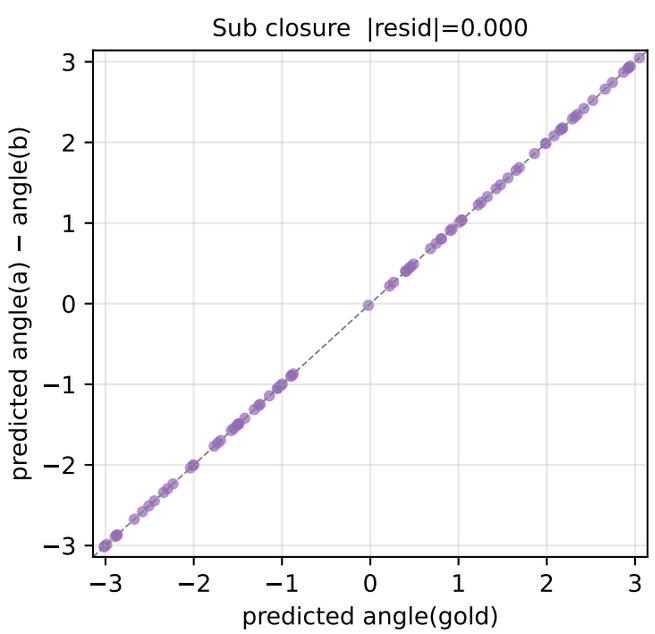

```
Period T = 50
Best cell:  step=6, layer=11
R²(a) = +0.954
cos-align(pred angle, true angle) = +0.989
max(operand)/T = 1.98
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  2.513 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

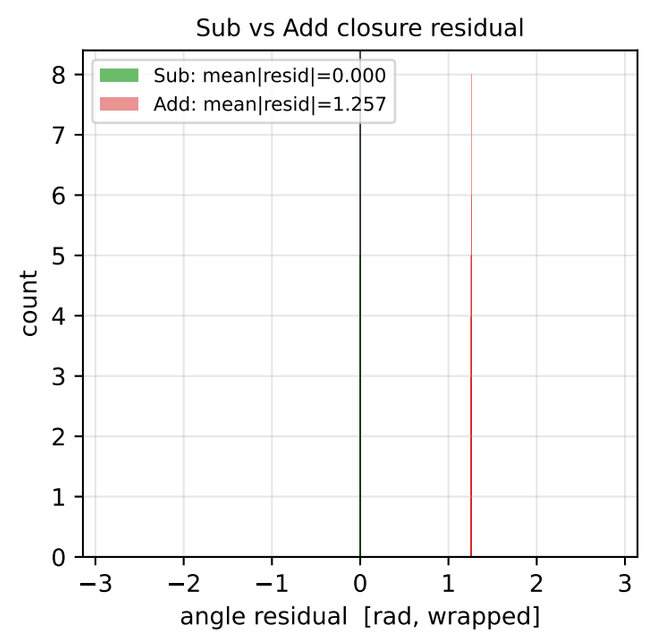
vary_a_2digit — Clock test at T=100





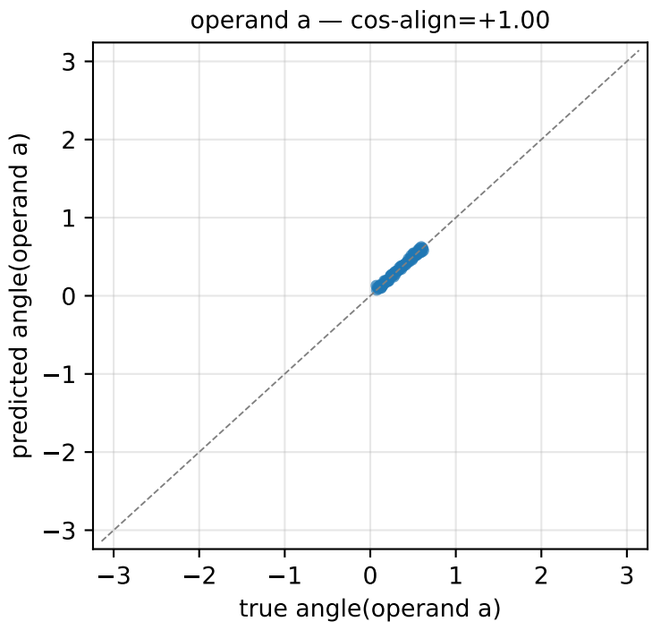


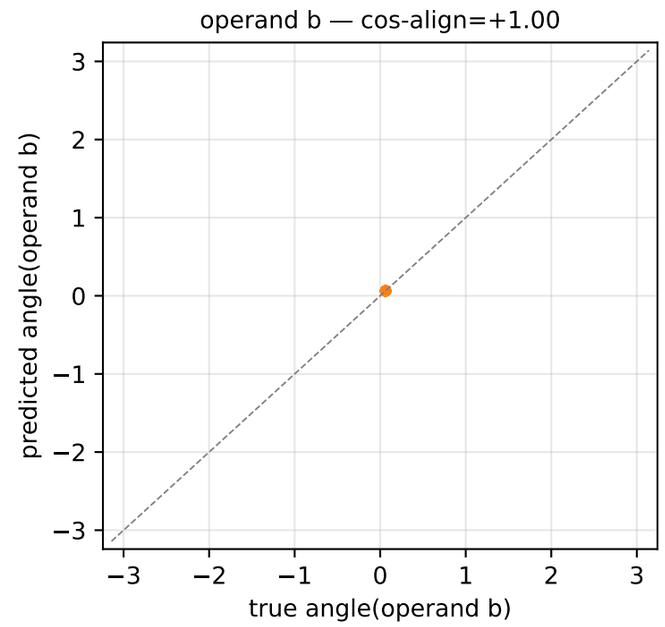


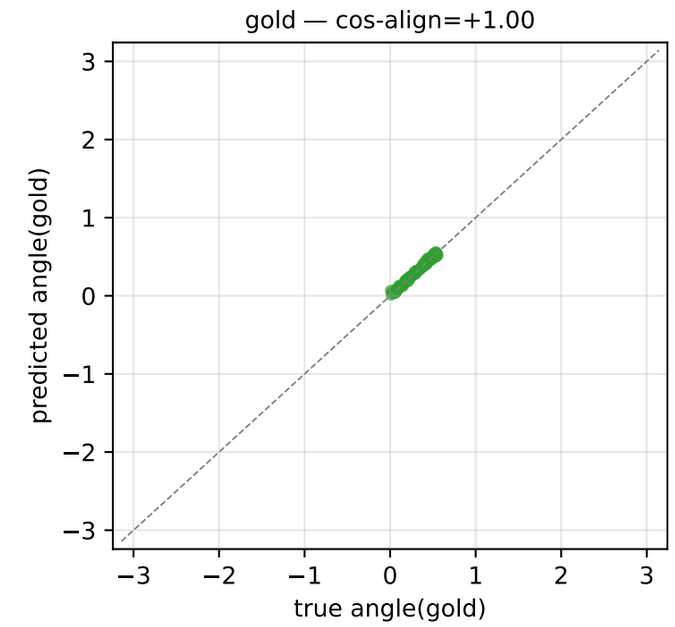


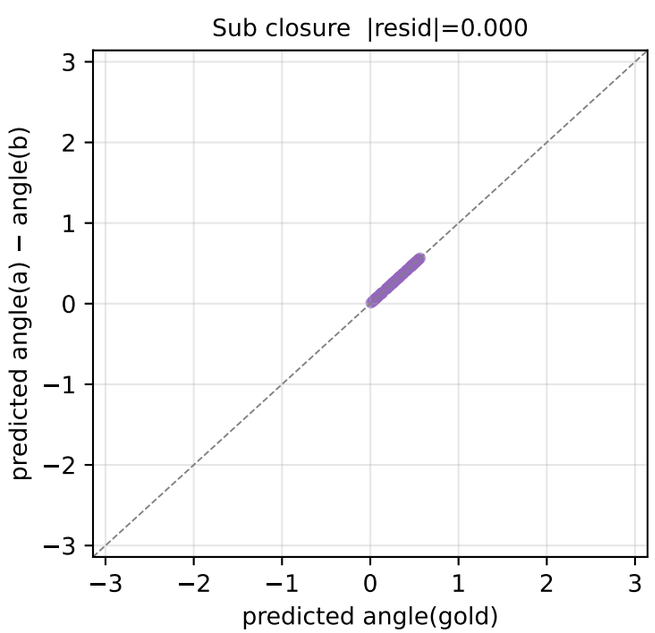

```
Period T = 100
Best cell:  step=6, layer=11
R²(a) = +0.971
cos-align(pred angle, true angle) = +0.993
max(operand)/T = 0.99
  Non-trivial Clock test? NO (near-linear)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  1.257 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

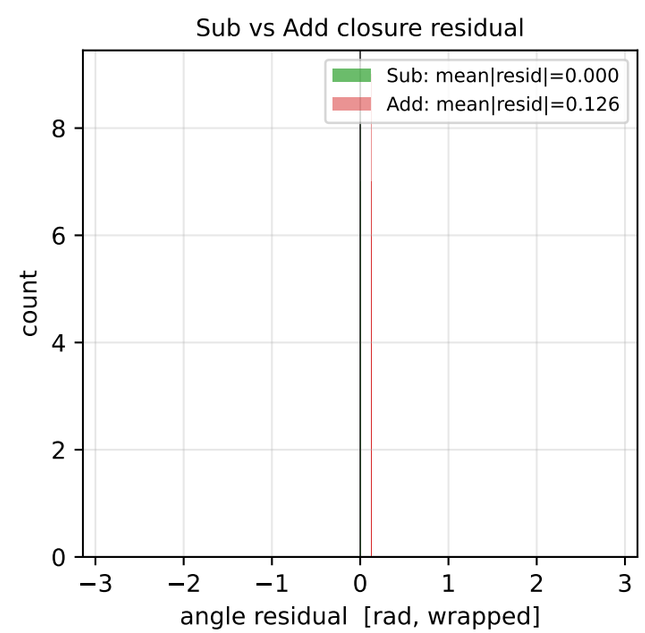
vary_a_2digit — Clock test at T=1000



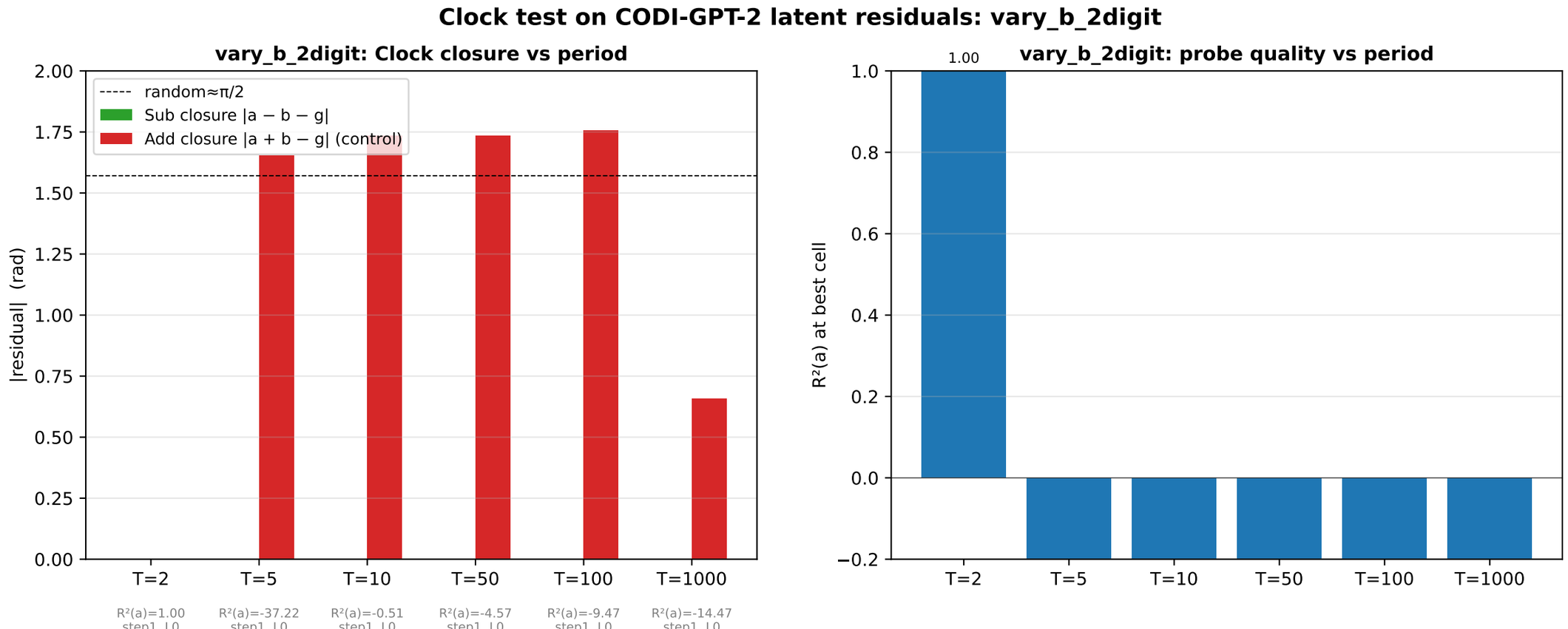




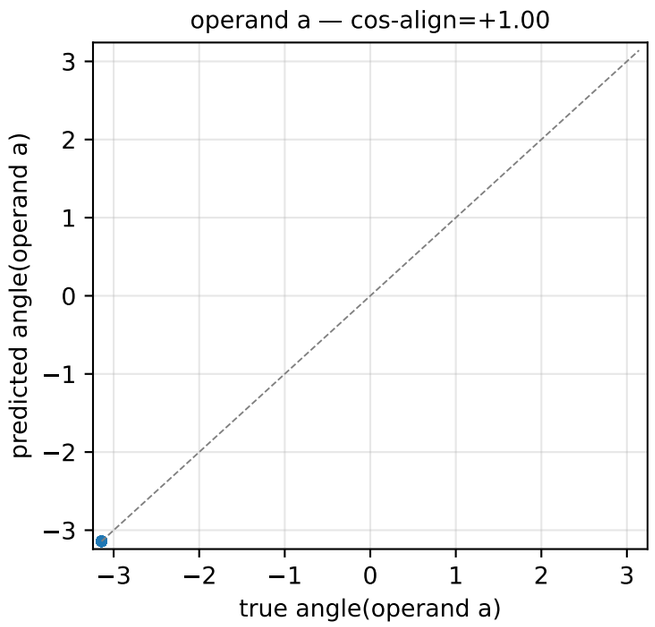


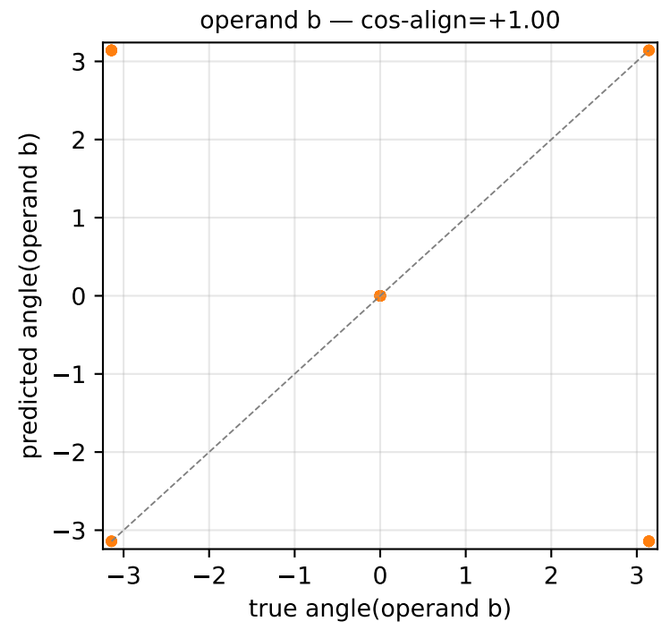


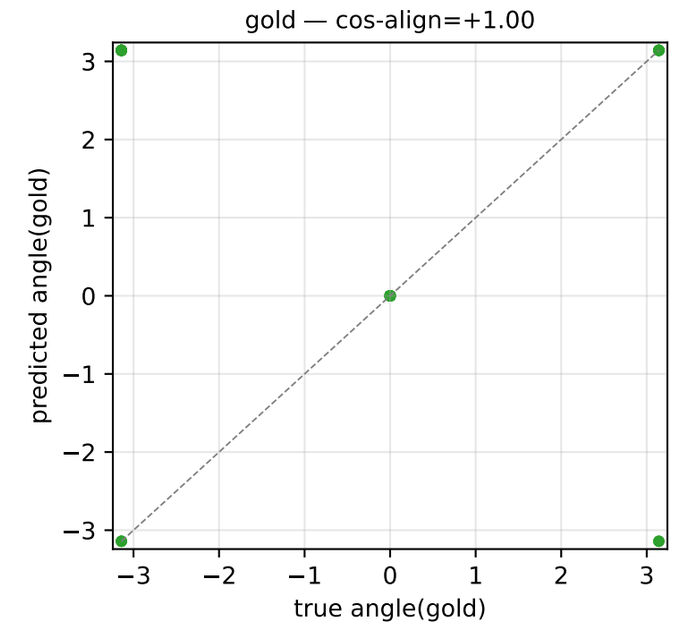

```
Period T = 1000
Best cell:  step=6, layer=10
R²(a) = +0.983
cos-align(pred angle, true angle) = +1.000
max(operand)/T = 0.10
  Non-trivial Clock test? NO (near-linear)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  0.126 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

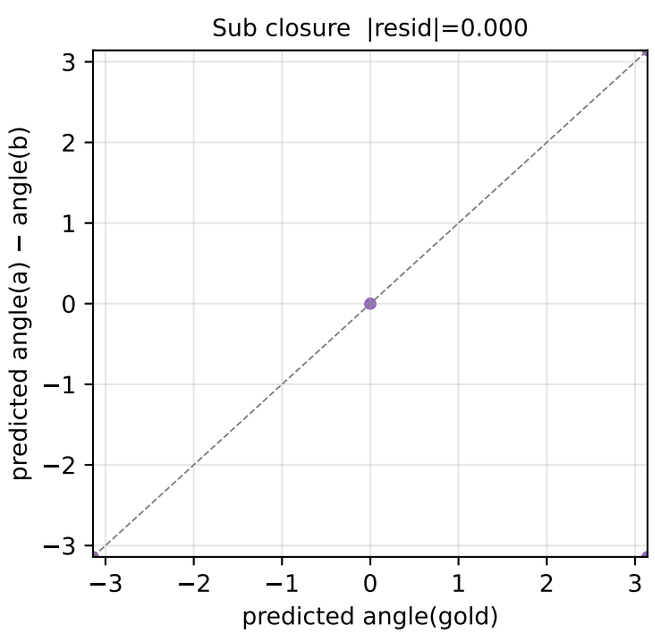

Clock test on CODI-GPT-2 latent residuals: vary_b_2digit
vary_b_2digit: Clock closure vs period
random≈π/2
Sub closure |a − b − g|
Add closure |a + b − g| (control)
|residual| (rad)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
T=2
T=5
T=10
T=50
T=100
T=1000
R²(a)=1.00
R²(a)=-37.22
R²(a)=-0.51
R²(a)=-4.57
R²(a)=-9.47
R²(a)=-14.47
vary_b_2digit: probe quality vs period
1.00
R²(a) at best cell
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
T=2
T=5
T=10
T=50
T=100
T=1000

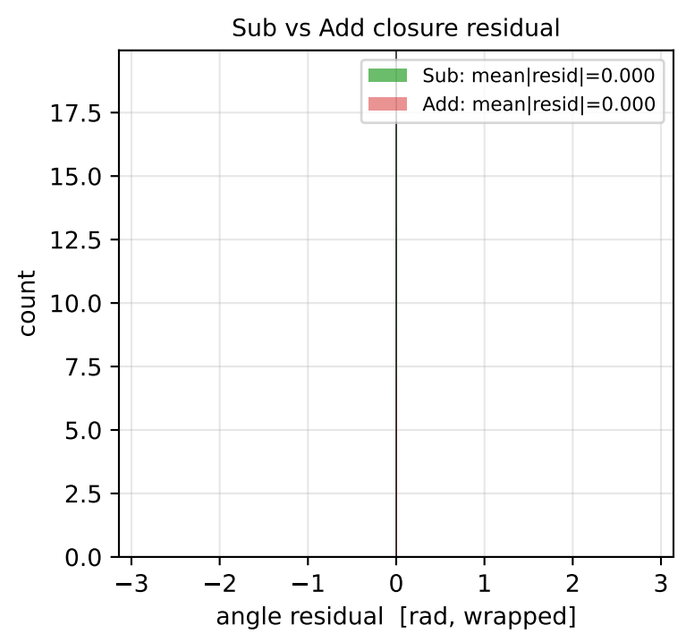
vary_b_2digit — Clock test at T=2





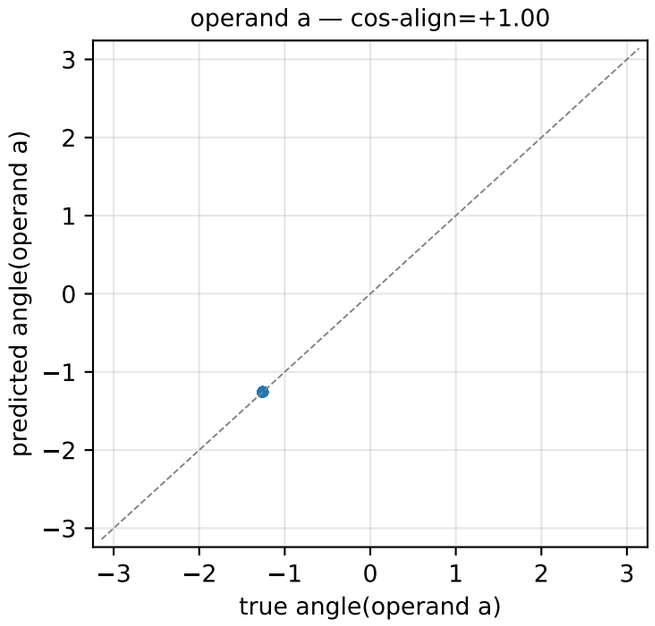


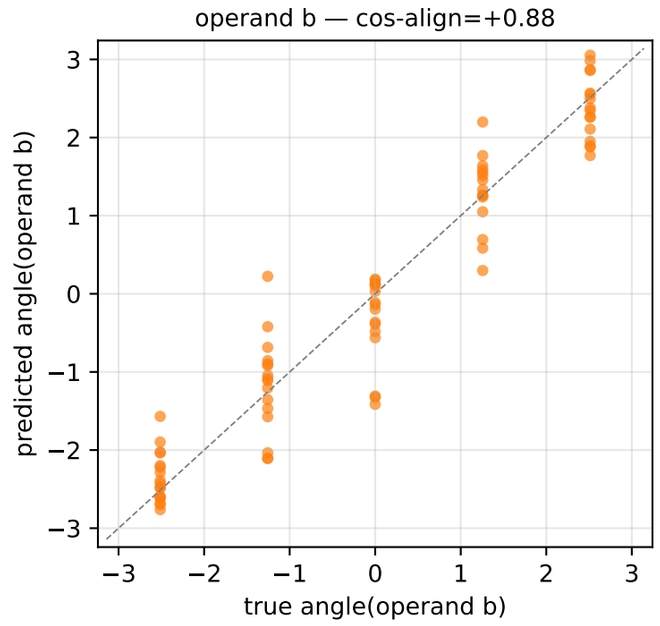


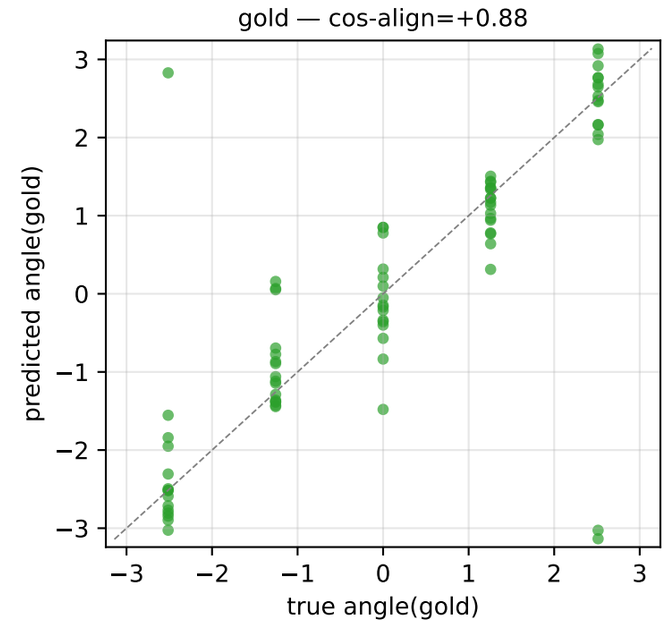


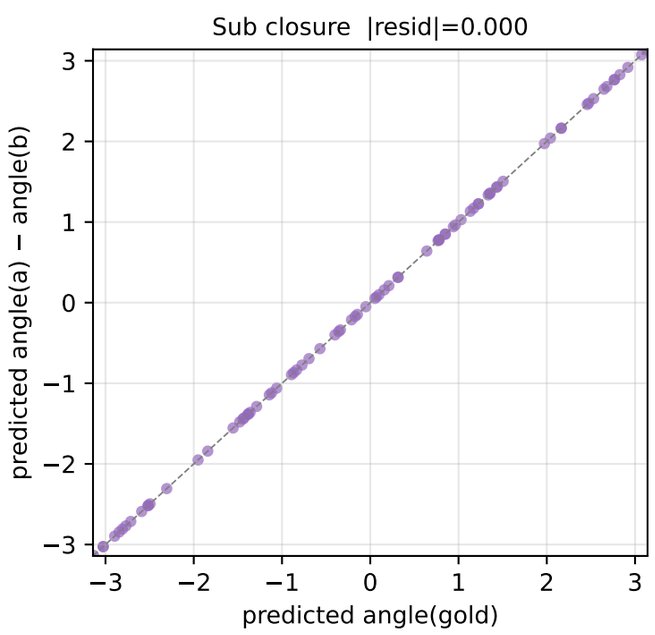

```
Period T = 2
Best cell:  step=1, layer=0
R²(a) = +1.000
cos-align(pred angle, true angle) = +1.000
max(operand)/T = 49.50
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  0.000 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

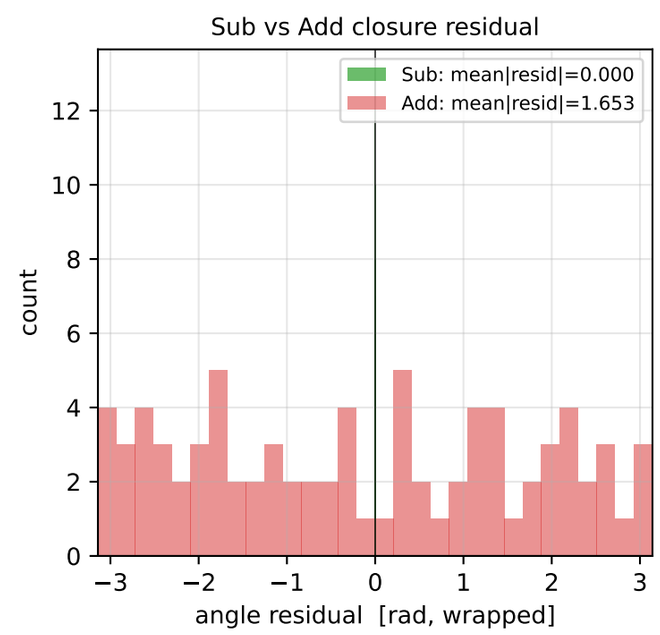
vary_b_2digit — Clock test at T=5





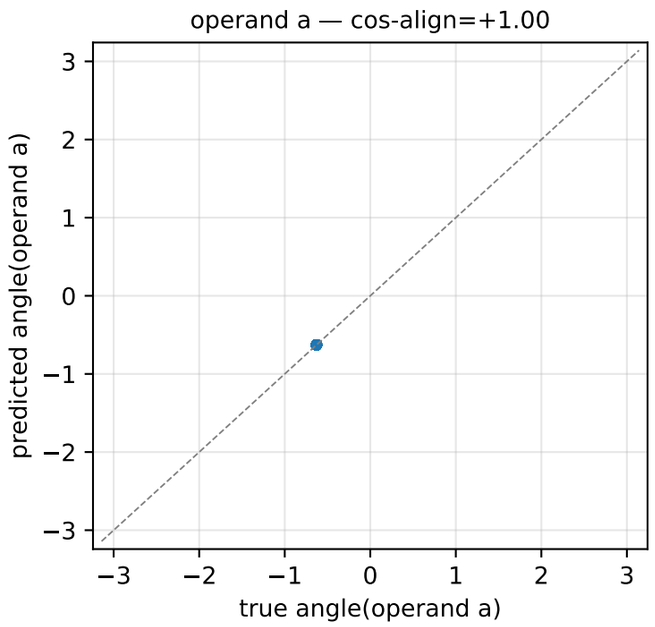


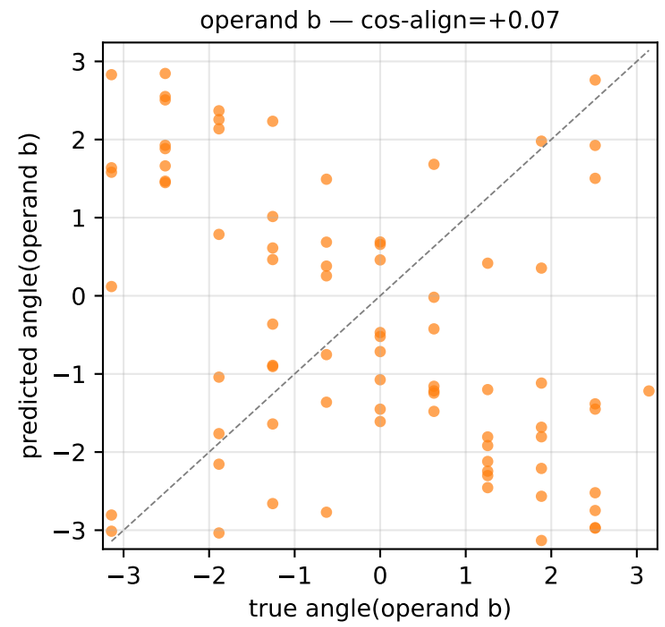


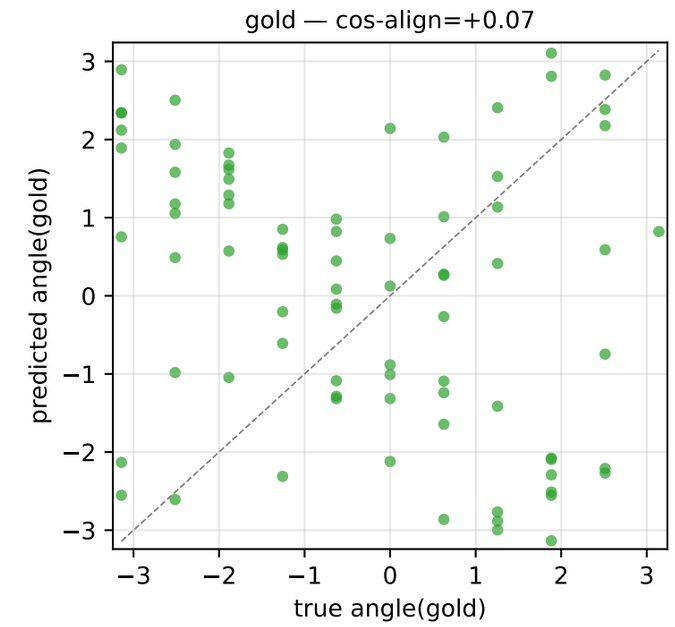


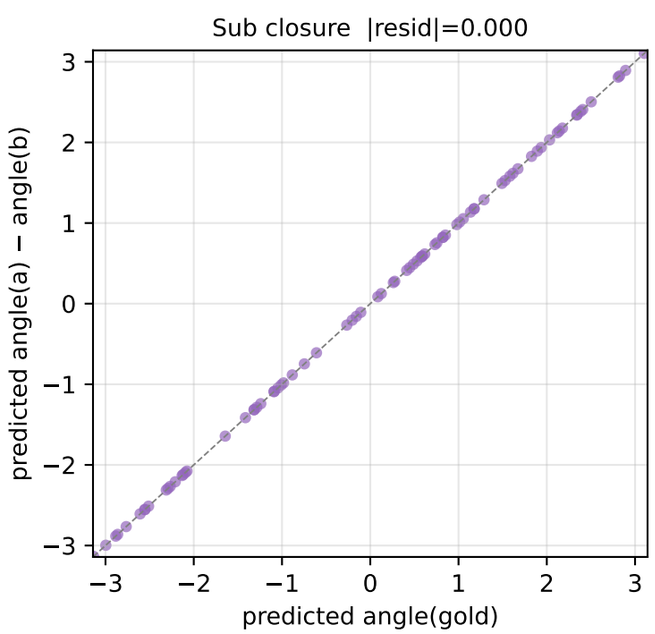

```
Period T = 5
Best cell:  step=1, layer=0
R²(a) = -37.222
cos-align(pred angle, true angle) = +1.000
max(operand)/T = 19.80
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  1.653 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

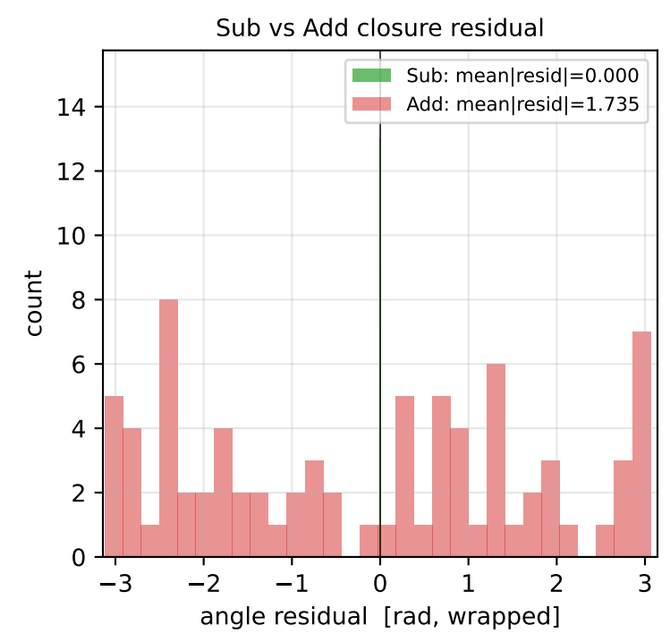
vary_b_2digit — Clock test at T=10





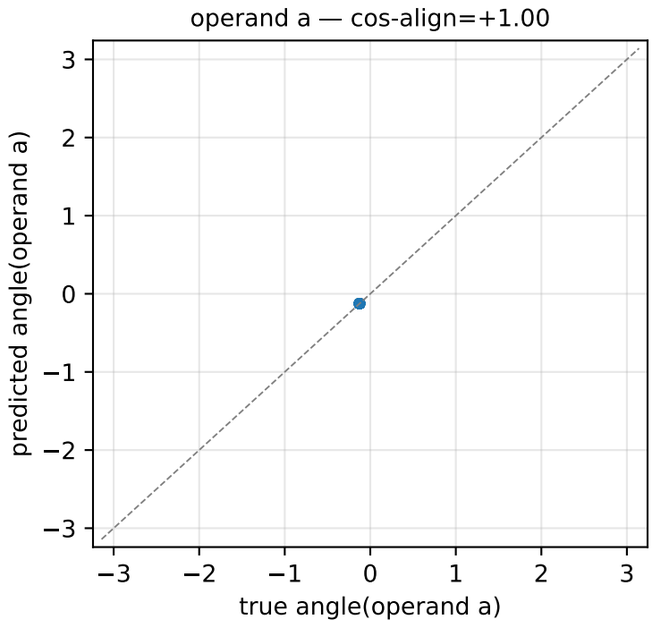


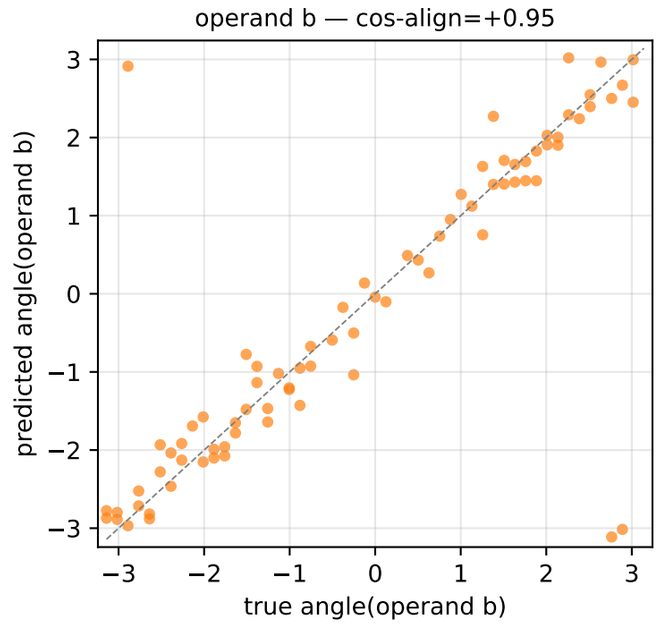


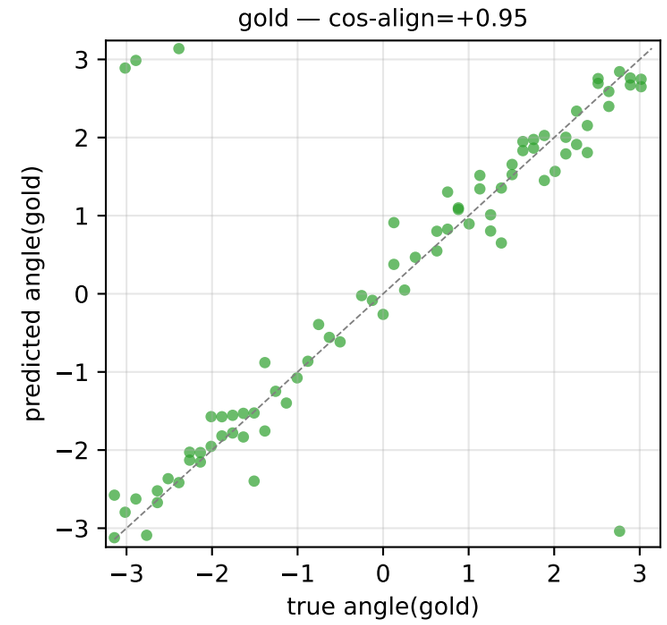


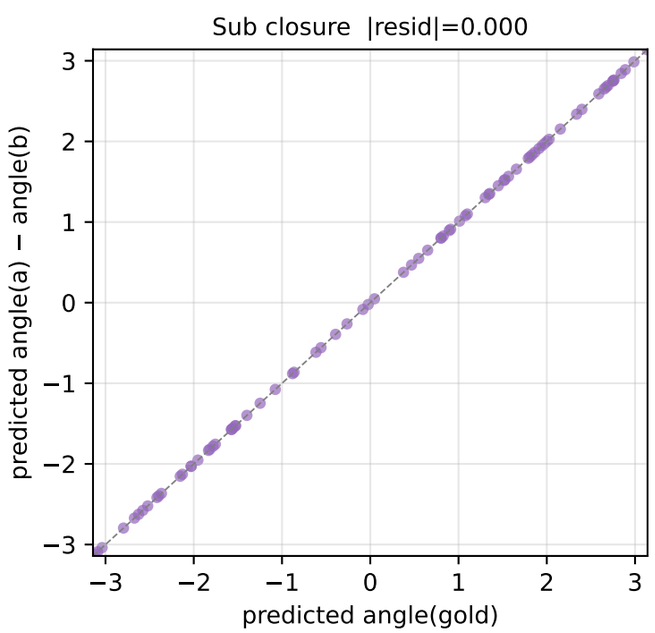

```
Period T = 10
Best cell:  step=1, layer=0
R²(a) = -0.509
cos-align(pred angle, true angle) = +1.000
max(operand)/T = 9.90
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  1.735 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

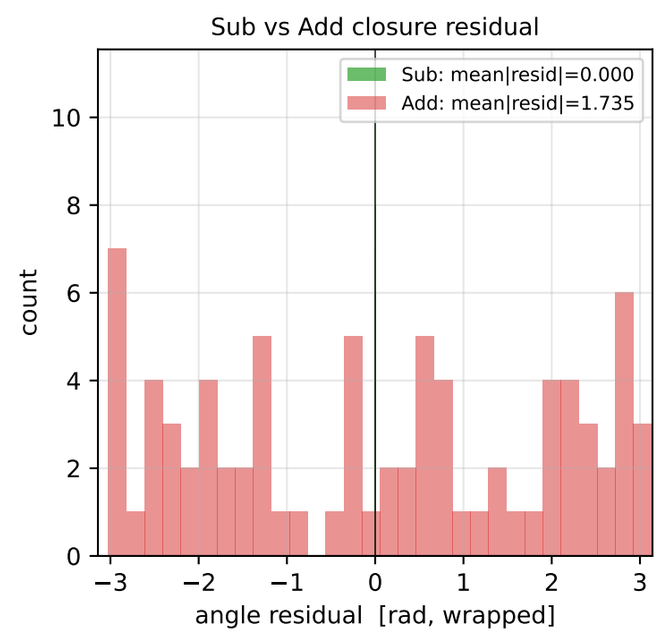
vary_b_2digit — Clock test at T=50





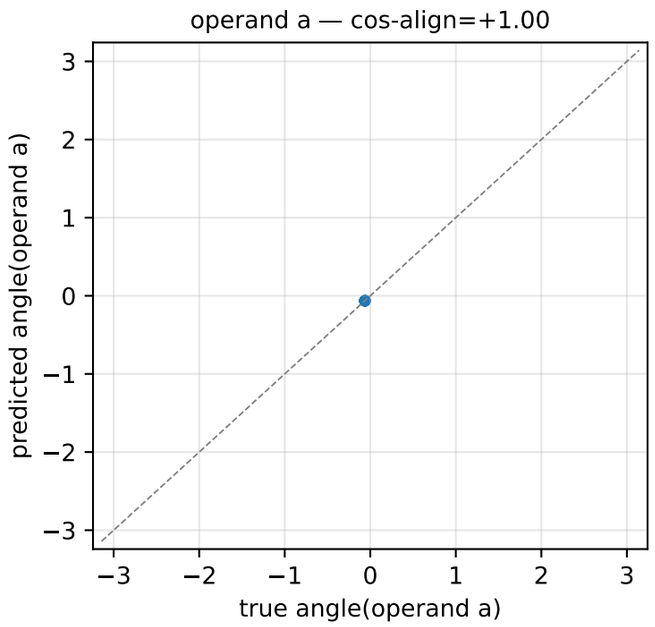


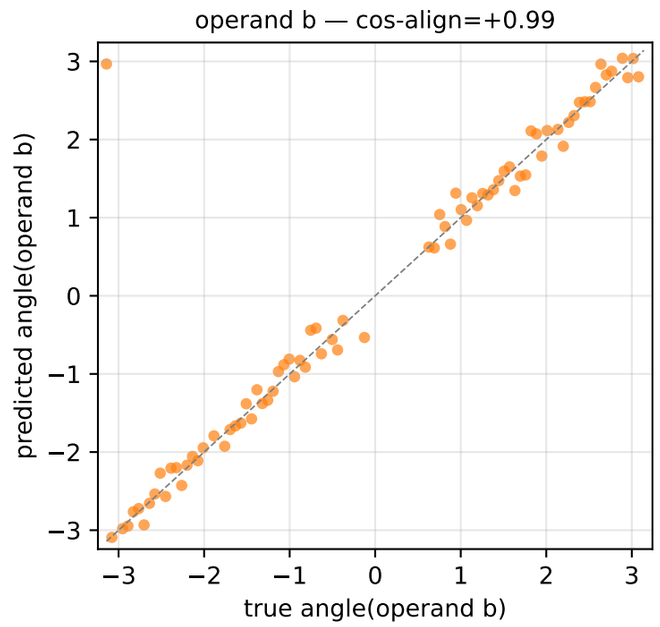


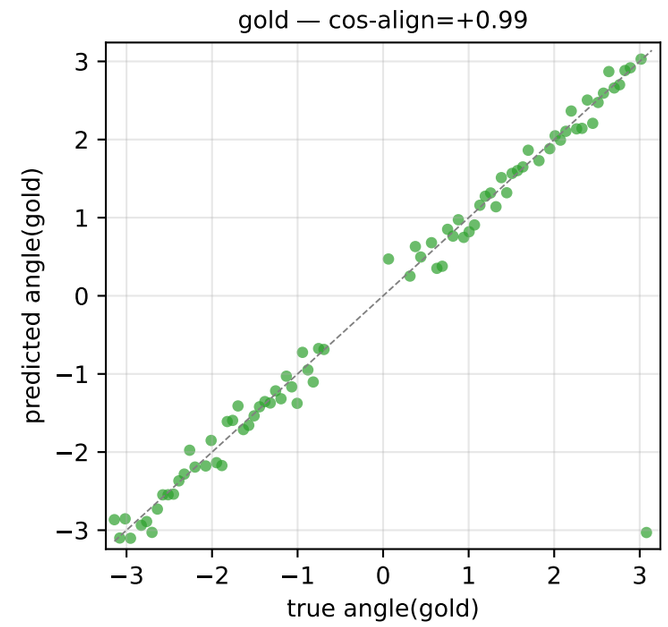


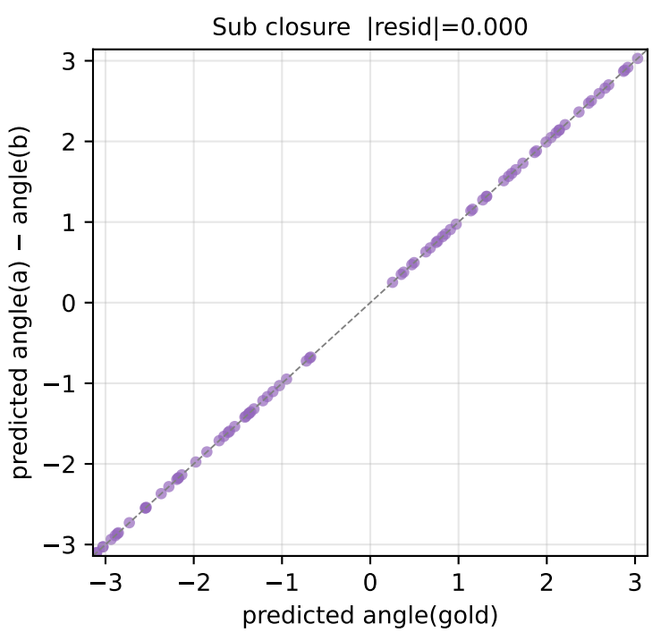

```
Period T = 50
Best cell:  step=1, layer=0
R²(a) = -4.569
cos-align(pred angle, true angle) = +1.000
max(operand)/T = 1.98
  Non-trivial Clock test? YES (operands wrap)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  1.735 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: STRONG (likely Clock-consistent)

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

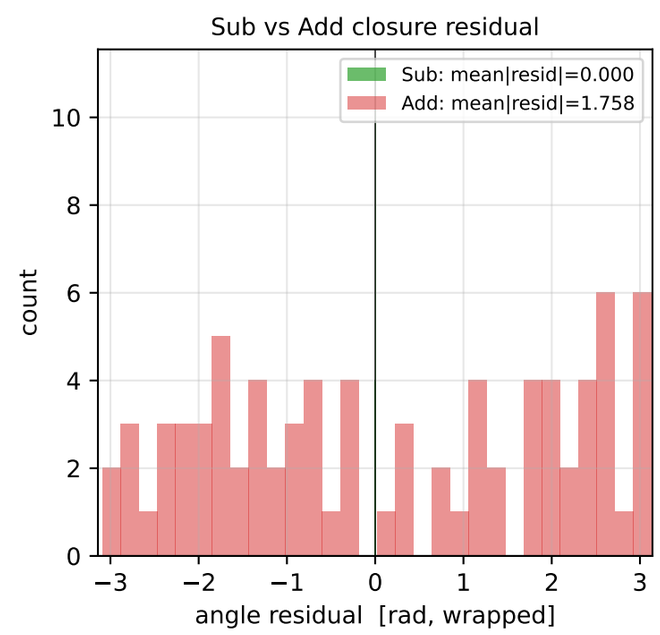
vary_b_2digit — Clock test at T=100





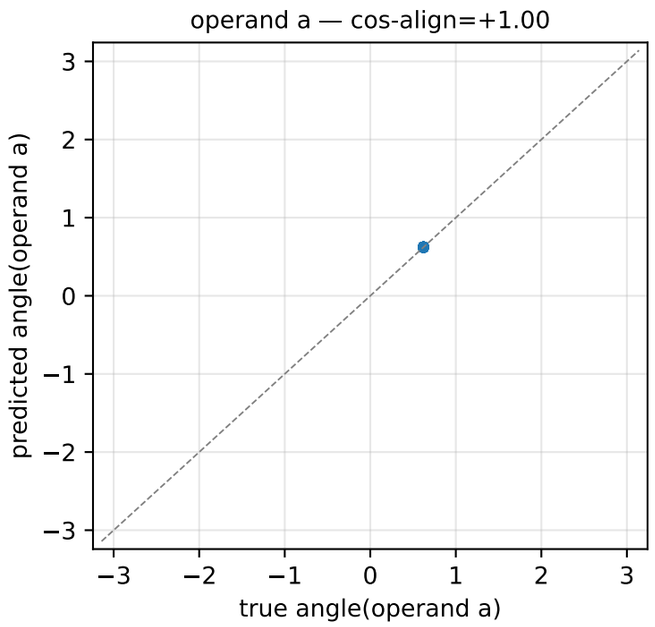


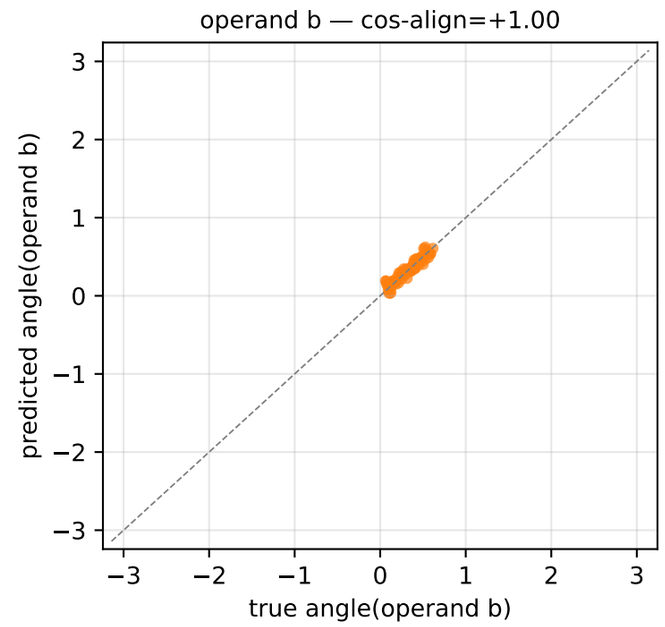


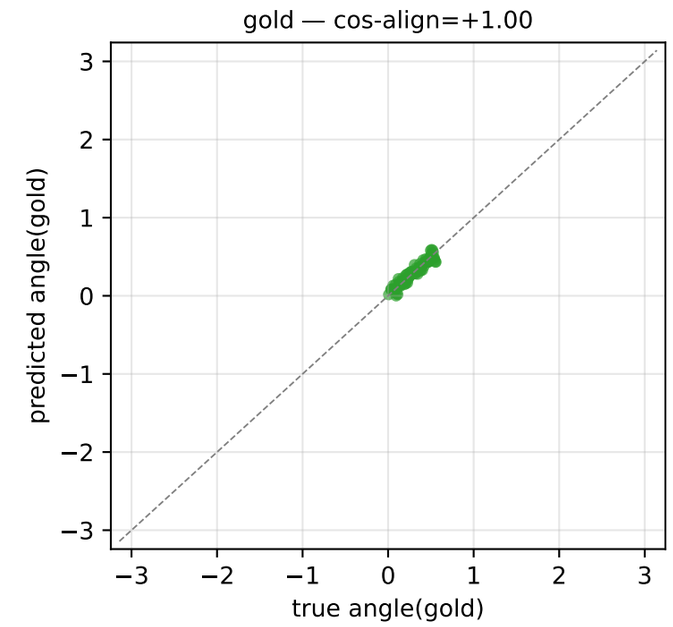


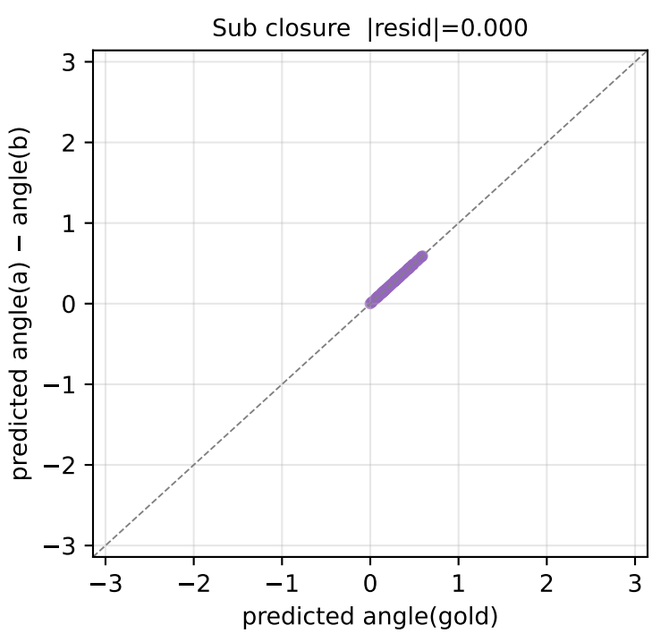

```
Period T = 100
Best cell:  step=1, layer=0
R²(a) = -9.469
cos-align(pred angle, true angle) = +1.000
max(operand)/T = 0.99
  Non-trivial Clock test? NO (near-linear)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  1.758 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```

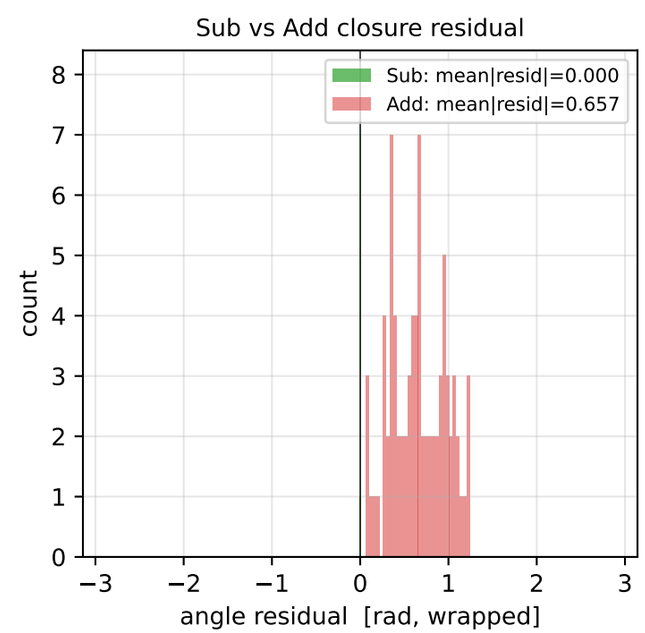
vary_b_2digit — Clock test at T=1000












```
Period T = 1000
Best cell:  step=1, layer=0
R²(a) = -14.474
cos-align(pred angle, true angle) = +1.000
max(operand)/T = 0.10
  Non-trivial Clock test? NO (near-linear)

|angle(a) − angle(b) − angle(gold)|  =  0.000 rad
|angle(a) + angle(b) − angle(gold)|  =  0.657 rad  (control)
random baseline ≈ π/2 ≈ 1.57 rad

Verdict: MODERATE

Note: this consistency test cannot distinguish
between a true helical Clock geometry and
independent linear encoding of a, b, gold +
arithmetic.  See clock_subspace_rank.py and
clock_causal_rotation.py for that.
```